GEORG-AUGUST-UNIVERSITÄT GÖTTINGEN IN PUBLICA COMMODA SEIT 1737

Fakultät für Physik

Master's Thesis

Improving the Reconstruction of Top-Quark Pairs in Dileptonic Final States in ATLAS Using Machine Learning

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ReferenceNumber: II.Physik-UniGö-MSc

Thesis period: 7th April 2025 until 30th June 2026

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Abstract

Many of the recent highlights in particle physics research are related to top-quark physics. These include both the stringent tests of spin correlations and quantum effects in pairs of top quarks ($t\bar{t}$), and the observation of a possible quasi-bound state resonance in the $t\bar{t}$ invariant mass spectrum. The latter presents a very exciting opportunity to extend our current understanding of non-relativistic Quantum Chromodynamics. Both effects are predominantly studied in the dilepton decay channel of the top quark in a mass range close to the $t\bar{t}$ production threshold.

The dilepton final state contains two neutrinos, besides two charged leptons and two b-quark jets. Probing this system requires a precise reconstruction of the top quarks, which is complicated by the presence of the two neutrinos. While analytical regression strategies primarily focus on inferring the neutrino momenta, the problem of correctly assigning b-jets to their parent top quarks remains largely understudied. However, many of the sensitive variables used in $t\bar{t}$ precision measurements depend critically on the correct assignment of the jets.

Inspired by the success of machine learning architectures in tackling the assignment challenge for hadronic decay channels, this work investigates using a transformer model for the dilepton channel. An architecture specifically tailored to this channel's topology is shown to outperform all existing methods in assigning reconstructed particles to their parent partons, improving the resolution of relevant physics observables. Furthermore, it is investigated how these efforts can be combined with neutrino regression methods to offer a full reconstruction pipeline. Applying these methods to existing and upcoming analyses promises further enhancements in the sensitivity and precision of searches and measurements alike.

Keywords: Top Quark, Dileptonic Decay, Event Reconstruction, Machine Learning, Transformer Networks

Contents

1. Introduction

The top quark, being the heaviest known elementary particle, plays a crucial role in the Standard Model of particle physics. Due to its large mass, it stands out in two ways; for one, as having the highest Yukawa-coupling to the Higgs boson and secondly, as being the shortest lived elementary particle. While, the Yukawa-coupling makes it very interesting for Higgs physics, the short lifetime of the top quark presents it as the only window allowing us to look at the bare coupling of quarks. The top quark decays almost exclusively into a W boson and a b-quark. These properties make processes involving top quarks as some of the most studied at the Large Hadron Collider (LHC) at CERN.
Top quarks are predominantly produced in pairs ($t\bar{t}$) via the strong interaction in proton-proton collisions at the LHC. Each W boson coming from the top decay can decay either leptonically (into a charged lepton and neutrino) or hadronically (into a pair of quarks). The dileptonic decay channel, where both W bosons decay leptonically, results in a final state with two charged leptons, two b-jets, and two neutrinos. This channel, while having a lower branching ratio compared to other decay modes, offers a cleaner experimental signature due to the presence of leptons and reduced hadronic activity. Leptons in particular can be reconstructed with a high efficiency in a hadron collider environment. However, the presence of two neutrinos in the final state poses significant challenges for event reconstruction, as they evade detection, leading to missing transverse energy ($\not{E}_{\mathrm{T}}$) in the event.
Due to its clean signature and sensitivity to various top-quark properties, the dileptonic $t\bar{t}$ channel is widely used for precision measurements of the top properties and spin correlations in particular. Two recent highlights in top-quark physics are the tests of spin correlations and quantum entanglement in pairs of top quarks [1], and the observation of a possible quasi-bound state resonance in the $t\bar{t}$ invariant mass spectrum [2, 3]. Both effects are predominantly studied in the dilepton decay channel of the top quark in a mass range close to the on-shell production threshold of two top quarks $m(t\bar{t}) \gtrsim 2 \cdot m_t \approx 350\,\mathrm{GeV}$. Compared to hadronic decays, the leptons allow a more accurate probing of the top quarks spin by preserving a larger fraction of the spin information [4, 5].
Both analyses rely on measurements of angular distributions and correlations between the decay products of the top quarks. Probing this system requires a precise reconstruction of the top quarks from their decay products, which is complicated by the presence of the two neutrinos. While analytical neutrino regression strategies have been studied extensively, the problem of correctly assigning b-jets to their parent top quarks remains largely unstudied. In channels with hadronically decaying top quarks, various methods for jet assignment have been developed and employed successfully [6]. However, in the dileptonic channel, the jet assignment problem was often simplified or overlooked due to a focus on neutrino reconstruction. However, many of the sensitive variables used in measurements of spin correlations depend critically on the correct assignment of the jets.

Misassignments degrade the resolution of reconstructed observables, ultimately limiting the precision of measurements.
Furthermore, the jet assignment problem is not only relevant for the reconstruction of the top quarks but also for the neutrino regression itself. Many of the conventional neutrino regression methods rely on kinematic constraints that are applied to the decay products of each top quark. If the jets are misassigned, these constraints can lead to incorrect solutions for the neutrino momenta, further degrading the reconstruction performance.
This thesis, seeks to address the jet assignment problem in close combination with the neutrino regression, with the goal of improving the overall reconstruction of dileptonic $t\bar{t}$ events. For this purpose, a novel jet assignment method based on transformer networks is developed and evaluated. To address the interdependence of jet assignment and neutrino regression, a model that performs both tasks simultaneously is developed and evaluated. The performance of the proposed methods is compared to conventional approaches, and their impact on the reconstruction of top quark properties, such as spin correlations, is assessed.
Begging with a review of the theoretical background Chapter 2 and the experimental setup in Chapter 3, the thesis provides the necessary context for understanding the challenges of reconstructing dileptonic $t\bar{t}$ events. The following Chapter 4 introduces the foundation of the machine learning techniques employed in this work. The general setup of $t\bar{t}$ analyses and the conventional reconstruction methods are outlined in Chapter 5. Starting with Chapter 6, the novel machine learning-based reconstruction methods are developed and evaluated. Following this, the performance of the proposed methods is assessed in Chapter 7, and they are applied to a physics analysis in Chapter 8 to demonstrate their in-situ performance and highlight their availability for future analyses. Finally, Chapter 9 summarizes the findings of this thesis and discusses potential future directions for improving the reconstruction of dileptonic $t\bar{t}$ events.

2. Theoretical Background

2.1. The Standard Model of Particle Physics

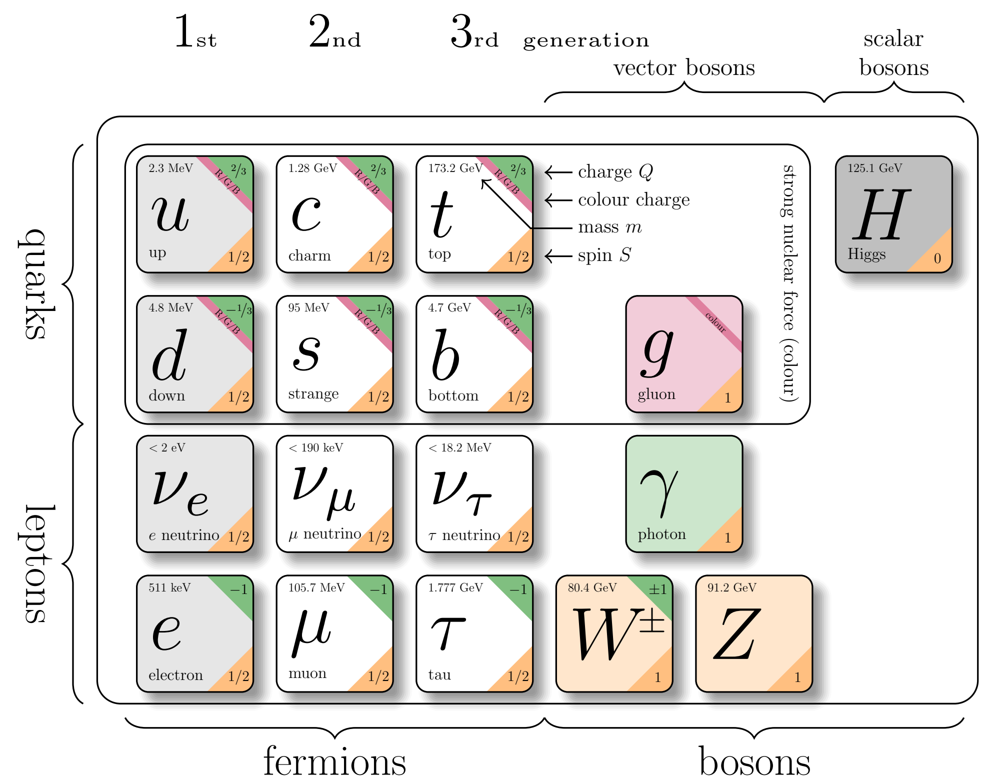


Figure 2.1.: The particles of the Standard Model. Particle properties taken from Reference [7].

The Standard Model (SM) of particle physics is a quantum field theory describing three of the four fundamental interactions of nature—electromagnetic, weak and strong—and the elementary particles on which they act. It is formulated as a renormalizable, Lorentz-invariant gauge theory with gauge group

$$SU(3)_C \times SU(2)_L \times U(1)_Y,$$

and its structure has been established in a series of seminal works [8–13]. A summary of the particle content is shown in Figure 2.1.
The elementary particles fall into two categories, fermions and bosons. Fermions are spin-$\frac{1}{2}$ fields that constitute matter, organised into three generations of quarks and leptons.

Quarks carry colour charge and participate in the strong interaction; leptons do not. Bosons are integer-spin gauge fields mediating the fundamental forces: the photon for electromagnetism, the $W^{\pm}$ and Z bosons for the weak interaction, and eight gluons for the strong interaction. The Higgs boson completes the SM and is responsible for electroweak symmetry breaking (EWSB), giving mass to the weak gauge bosons and, through Yukawa couplings, to the fermions. The SM Lagrangian can be written schematically as

$$\mathcal{L}_{\mathrm{SM}} = \mathcal{L}_{\mathrm{gauge}} + \mathcal{L}_{\mathrm{fermion}} + \mathcal{L}_{\mathrm{Yukawa}} + \mathcal{L}_{\mathrm{Higgs}},$$

where the first term contains the gauge kinetic terms and self-interactions, the second describes the fermions and gauge covariant derivatives, the third encodes the Yukawa couplings to the Higgs field, and the last term contains the Higgs potential and EWSB mechanism.
Below, the three interaction sectors are briefly summarised, with emphasis on the aspects relevant for top-quark physics and LHC phenomenology.

2.1.1. Electromagnetic Interaction

The electromagnetic interaction is described by quantum electrodynamics (QED), the $U(1)_{\mathrm{EM}}$ gauge theory that remains after EWSB. Its massless gauge boson, the photon, couples to electric charge and does not self-interact. Because QED is an Abelian theory, its coupling runs only mildly with energy and remains perturbative at all experimentally accessible scales.

2.1.2. Weak Interaction

The weak interaction forms, together with electromagnetism, the electroweak theory based on the gauge group $SU(2)_L \times U(1)_Y$. The vacuum expectation value of the Higgs field spontaneously breaks the symmetry to $U(1)_{\mathrm{EM}}$, giving masses to the weak gauge bosons through the Higgs mechanism [14–16]. The resulting massive $W^{\pm}$ and Z bosons mediate a short-range force.
A defining property of the weak interaction is its *chiral* structure: charged-current interactions couple only to left-handed fermions and right-handed antifermions. This feature, predicted in [17] and confirmed in the Wu experiment [18], leads to maximal parity violation and characteristic angular distributions of weak decay products. The vertex factor of the charged-current interaction is given by

$$\frac{-ig}{2\sqrt{2}}\gamma^{\mu}(1-\gamma^{5}), \tag{2.1}$$

where g is the weak coupling constant, γ^{μ} are the Dirac matrices. The projection operator $(1-\gamma^{5})$ leads to a coupling exclusively to the left-handed components. Based on this, the left-handed particles are arranged in $SU(2)_L$ doublets, while the right-handed particles are singlets.
For quarks, the weak eigenstates d', s', b' are not identical to their mass eigenstates d, s, b. This misalignment is described by the Cabibbo-Kobayashi-Maskawa (CKM) matrix [19,

20], which enables flavour-changing coupling in charged-current interactions.

2.1.3. Strong Interaction

The strong interaction is described by quantum chromodynamics (QCD), a non-Abelian gauge theory with gauge group $SU(3)_C$. Quarks carry colour charge, while gluons carry a colour and anticolour index, giving rise to self-interactions through three- and four-gluon vertices. These features lead to two fundamental properties:

Asymptotic freedom. The QCD coupling decreases at high momentum transfer [11, 21], allowing perturbative calculations of hard processes such as top-quark pair production. In the high-energy regime of the LHC, quarks and gluons effectively behave as quasi-free particles.

Confinement. At low energies, the QCD coupling becomes large, preventing coloured states from existing in isolation. Instead, they form colour-neutral bound states—hadrons. The transition from short-distance partons to long-distance hadrons involves nonperturbative dynamics encapsulated through hadronisation models.
A crucial consequence for collider physics is that quarks and gluons produced in hard interactions undergo a cascade of QCD emissions before hadronising into jets. This process is typically described in terms of:

- **Parton distribution functions (PDFs)**, characterising the proton structure and entering via the QCD factorisation theorem [22].
- **Initial- and final-state radiation (ISR/FSR)**, arising from soft and collinear gluon emission.
- **Parton showers**, which resum logarithmically enhanced emissions and are implemented in Monte Carlo generators such as PYTHIA, first developed in [23, 24] and extended in modern formulations [25].
- **Hadronisation models**, such as the Lund string model [26, 27], which describe the confinement-driven formation of hadrons.

These aspects of QCD are essential for the reconstruction of hadronic final states and for Monte Carlo simulations of LHC physics.

2.2. Top-Quark Physics

The top quark is the heaviest known elementary particle. The prediction of its existence arose due the already observed CP-Violation in the kaon system, which required a third generation of quarks to be explained within the SM framework [20]. Furthermore, the discovery of the bottom quark in 1977 [28] implied the existence of its weak isospin partner, the top quark. The top quark was eventually discovered in 1995 by the CDF and DØ

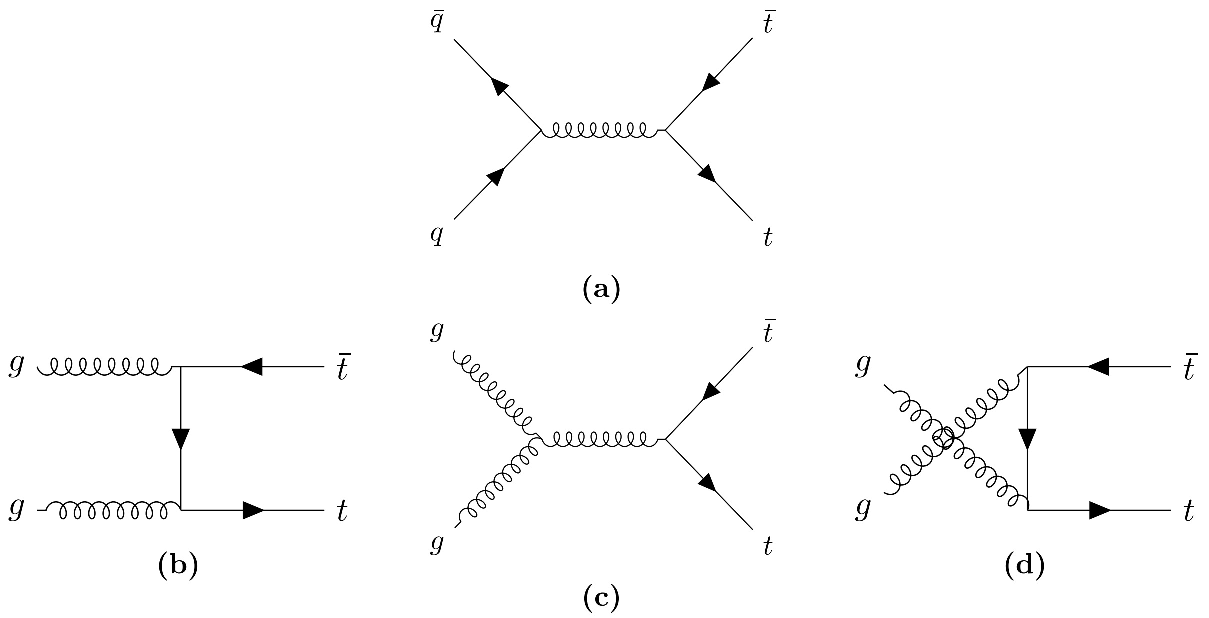


Figure 2.2.: Leading order Feynman diagrams for top-quark pair production in (a) quark-antiquark annihilation and (b), (c) & (d) gluon fusion (ggF).

collaborations at the TEVATRON collider [29, 30]. Currently, the world average of the top-quark mass is measured to be [7]

$$m_t = 172.56 \pm 0.31\,\text{GeV}. \tag{2.2}$$

Along with the largest mass, the top quark also has the strongest Yukawa coupling to the Higgs boson, making it interesting for studies of the Higgs mechanism and electroweak symmetry breaking.

Due to its large mass, the top quark also has the shortest lifetime of all quarks, about 5×10^{-25} s [7], which is shorter than the typical hadronisation timescale of 3×10^{-24} s [31]. Therefore, the top is the only quark that decays before it hadronises, allowing for direct studies of its properties through its decay products.

Notably, the top quark also decays before its spin can decorrelate, preserving spin information in the angular distributions of its decay products [32]. This unique feature enables precise measurements of spin correlations and polarisation of top quarks.

At hadron colliders like the LHC, top quarks are predominantly produced in pairs via the strong interaction. The leading order (LO) Feynman diagrams for top-quark pair production are shown in Figure 2.2. At the LHC, the dominant production mechanism is gluon fusion (ggF). This is due to the high gluon density in the proton at the relevant Bjorken-x values for top-quark production.

The CKM matrix element V_{tb} is close to unity [7], leading to a nearly exclusive decay of the top quark into a W boson and a b quark. The W can subsequently decay either leptonically into a charged lepton and a neutrino, or hadronically into a pair of quarks, with branching ratios of about $\text{BR}(W \to \ell\nu) = 1/3$ and $\text{BR}(W \to q\bar{q}) = 2/3$, respectively [7].

Based on these W decay modes, top-quark pair events are categorised into three channels. In the **dileptonic channel**, both W bosons decay leptonically ($W^+ \to \ell^+\nu_\ell$, $W^- \to \ell^-\bar{\nu}_\ell$), resulting in two charged leptons, two neutrinos and two b quarks in the final state. This channel has a branching ratio of approximately 1/9. The **semileptonic channel** occurs when one W boson decays leptonically and the other hadronically ($W \to q\bar{q}'$), yielding one charged lepton, one neutrino, four quarks (including two b quarks) in the final state. This is the most common channel, with a branching ratio of about 4/9. Finally, in the **all-hadronic channel**, both W bosons decay hadronically, producing six quarks (including two b quarks) in the final state. This channel has the highest branching ratio of approximately 4/9 but suffers from large multijet background contamination in the busy environment of hadron colliders like the LHC.

2.2.1. Spin Correlations in Top-Quark Pairs

As elaborated upon before, the top quark decays before it can hadronise or its spin decorrelates. Therefore, the spin information of the top quark is directly transferred to its decay products. In top-quark pair production, the spins of the top and antitop quarks are correlated due to the production mechanism. Due to the chiral nature of the weak interaction, these spin correlations manifest in the angular distributions of the decay products, particularly the charged leptons in the dileptonic decay channel.
Recent studies at the LHC began to explore potential effects of quantum entanglement in top-quark pairs [1]. Entanglement generally describes a quantum mechanical phenomenon where the quantum states of two or more particles become correlated in such a way that the state of one particle cannot be described independently of the state of the other(s), even when the particles are separated by large distances [33]. This property is expressed as the system having a non-separable density matrix. While a general description of this phenomenon is beyond the scope of this thesis, entanglement can be tested using specific criterions. One such criterion is the *Peres-Horodecki criterion* [34, 35], which states that if the partial transpose of the density matrix of a bipartite system has at least one negative eigenvalue, then the system is entangled. Considering the top-quark pair's spin as a bipartite system of two spin-$\frac{1}{2}$ particles (qubits), this criterion can be applied to test for entanglement in their spin states.
In Reference [36] it was proposed to measure entanglement in top-quark pairs produced at the LHC by analysing the angular distributions of their decay products, specifically the charged leptons in the dileptonic decay channel. The study showed that by measuring the opening angle between the two leptons, one can construct an observable sensitive to the entanglement of the spin system of the top quarks. Due to the kinematics of the top-quark pair production, the entanglement is expected to be more pronounced in certain regions of phase space, one such region is at low invariant masses of the top-quark pair system. One particular region of interest is the so-called *threshold region*, where the top-quark pair is produced with low relative velocity. In this region, the top quark is produced as a spin singlet state in about $\sim 80\%$ of the cases, leading to a strong entanglement signature [37]. The ATLAS collaboration has recently reported the first experimental evidence of quantum correlations consistent with entangled spin quantum states in top-quark pairs [1]. Measurements of this variable, are particularly sensitive

in the dilepton channel, because, the lepton has the highest spin-analysing power [4, 5]. The spin-analysing power measures, to which extent the spin-information of the parent is imprinted in the angular distributions of the decay products. Due to QCD effects, this is lower for light jets compared to leptons.

2.2.2. Top-Quark Pair Qausi-Bound-State Effects

Even though the top quark decays before it can hadronise, near the production threshold of top-quark pairs, the strong interaction between the top and top antiquarks can lead to the formation of transient bound states, coined as toponium [37–39]. These bound state effects can influence the production cross section and kinematic distributions of top-quark pairs near threshold. The CMS collaboration has recently observed an excess in the invariant mass distribution of top-quark pairs near threshold, consistent with the presence of toponium bound state effects [3]. The modelling of these effects is based on non-relativistic QCD (NRQCD) calculations, which account for the strong interaction dynamics between the top and top antiquarks in the near-threshold region [40, 41]. Understanding and accurately modelling these bound state effects is crucial for precision measurements of top-quark properties and for searches for new physics in top-quark pair production.
This study is also investigating the threshold region of top-quark pair production and makes use of angular correlations between the decay products as a probe of the underlying dynamics, including potential bound state effects.

2.2.3. Dileptonic Decay Channel

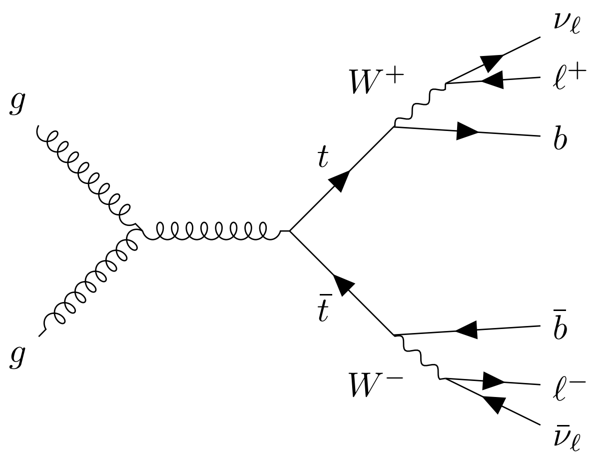


Figure 2.3.: Leading order Feynman diagram for top-quark pair production via gluon fusion with dileptonic decay.

The dileptonic decay channel, where both W bosons decay leptonically, has a branching ratio of about 1/9 and results in a final state with two charged leptons, two neutrinos and two b quarks, as illustrated in Figure 2.3. Due to the lower branching ratio, the dileptonic channel has a smaller event yield compared to the other channels. However, it offers a cleaner signature with reduced background contamination, making it particularly suitable for precision measurements of top-quark properties. The presence of two neutrinos in the final state leads to missing transverse energy ($\not{E}_{\mathrm{T}}$) in the event, complicating the full reconstruction of the top quark's kinematics. Advanced techniques, such as kinematic fitting and multivariate analyses, are often employed to address these challenges and extract maximum information from the dileptonic events.

Kinematic Constraints

Assuming both W bosons and top quarks are on-shell, the following kinematic constraints can be applied to the dileptonic decay channel,

$$
\begin{aligned}
(p_{\ell_1} + p_{\nu_1})^2 &= m_W^2, && (2.3)\\
(p_{\ell_2} + p_{\nu_2})^2 &= m_W^2, && (2.4)\\
(p_{b_1} + p_{\ell_1} + p_{\nu_1})^2 &= m_t^2, && (2.5)\\
(p_{b_2} + p_{\ell_2} + p_{\nu_2})^2 &= m_t^2, && (2.6)\\
(\vec{p}_{\nu_1} + \vec{p}_{\nu_2})_x &= \not{E}_{\mathrm{T}x}, && (2.7)\\
(\vec{p}_{\nu_1} + \vec{p}_{\nu_2})_y &= \not{E}_{\mathrm{T}y}. && (2.8)
\end{aligned}
$$

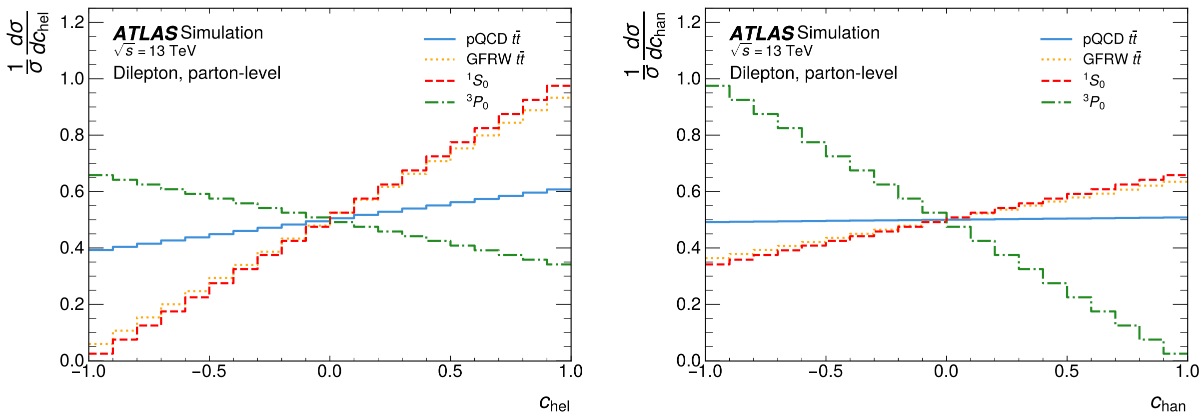
If one assumes the neutrinos to be massless, these six equations provide constraints on the six unknown components of the two neutrino momenta. Due to the algebraic nature of the constraints, multiple solutions can exist for a given event, leading to ambiguities in the reconstruction. Due to the effects of detector resolution and off-shell particles, this system of equation can also have no real-valued solution [42, 43]. Various methods have been developed to address these challenges, including likelihood-based approaches [44], matrix element methods, and machine learning techniques [45], which aim to optimally utilise the available information and improve the accuracy of the top quark's kinematic reconstruction in dileptonic events.

2.2.4. Spin-Sensitive Angular Variables

To study spin correlations and entanglement in top-quark pairs, a spin-sensitive angular variable is used. For the measurements of the quasi-bound-state effects, an additional angular variable is employed. Both variables are defined using the charged leptons from the dileptonic decay of the top-quark pair.
To define the variables, the momenta of the leptons are first boosted into their respective parent top-quark rest frames $\hat{\ell}_t^+$ and $\hat{\ell}_t^-$. The inner product of these unit vectors then defines the variable

$$c_{\mathrm{hel}} = \hat{\ell}_t^+ \cdot \hat{\ell}_t^- \,. \tag{2.9}$$

Analogously, one can define the variable c_{han}, where either of the lepton momenta components parallel to the parent top-quark momentum in the $t\bar{t}$ rest frame is flipped before taking the inner product [46, 47]. This variable adds important context by enabling isolation of scalar states.



(a) c_{hel} **(b)** c_{han}

Figure 2.4.: Simulation of the distributions of $c_{\text{hel}}$and $c_{\text{han}}$for dileptonic decaying top-quark pairs produced near threshold for different spin states and combinations thereof. The distributions are normalised to unity and at parton level. The Figures are taken from Reference [2].

3. Experimental Setup

3.1. The LHC

The Large Hadron Collider (LHC) at CERN is a synchrotron designed to accelerate protons and lead nuclei. This work focuses on the proton-proton collisions during Run 2. During Run 2 from 2015 to 2018, the LHC operated at a centre of mass energy of $\sqrt{s} = 13\,\text{TeV}$ with an integrated luminosity of $\mathcal{L}_{int} \approx 140\,\text{fb}^{-1}$ [48].
After completion of Run 3 in June 2026 and an operational pause, the LHC will operate with increased luminosity as the High-Luminosity Large Hadron Collider (HL-LHC). During this phase, it is planned to accumulate data with an integrated luminosity of around $\mathcal{L} = 3\,\text{ab}^{-1}$ [49].

3.2. The Detector

The ATLAS detector is a general-purpose particle detector at the LHC at CERN. It can roughly be separated into three elements: the inner detector, the calorimeter and the muon spectrometer [50].

3.2.1. The Coordinate System

The interaction point marks the origin of the ATLAS coordinate system. The z-axis is orientated along the beamline, the x-axis points towards the centre of the LHC and the y-axis points upwards.
Using this convention, several kinematic variables are defined. One of which is the transverse momentum defined as

$$p_T = \sqrt{p_x^2 + p_y^2} \tag{3.1}$$

Because of the cylindrical shape of the ATLAS detector and the cylindrical symmetry of the interactions, the azimuthal angle is used as another variable. The polar angle relative to the z-axis can be used to define the pseudorapidity which can also be expressed in terms of the momentum

$$\eta = -\ln\left(\tan\left(\frac{\theta}{2}\right)\right) = \ln\left(\frac{|\vec{p}| + p_z}{|\vec{p}| - p_z}\right) . \tag{3.2}$$

In the ultra-relativistic limit $m \ll |\vec{p}|$ the pseudorapidity is equal to the rapidity defined as

$$y = \frac{1}{2} \ln \left(\frac{E + p_z}{E - p_z} \right) . \tag{3.3}$$

These variables are convenient for the usage in the ATLAS experiment because p_T, ϕ, and differences of η are invariant under Lorentz-boost along the z-axis.

3.2.2. Inner Detector

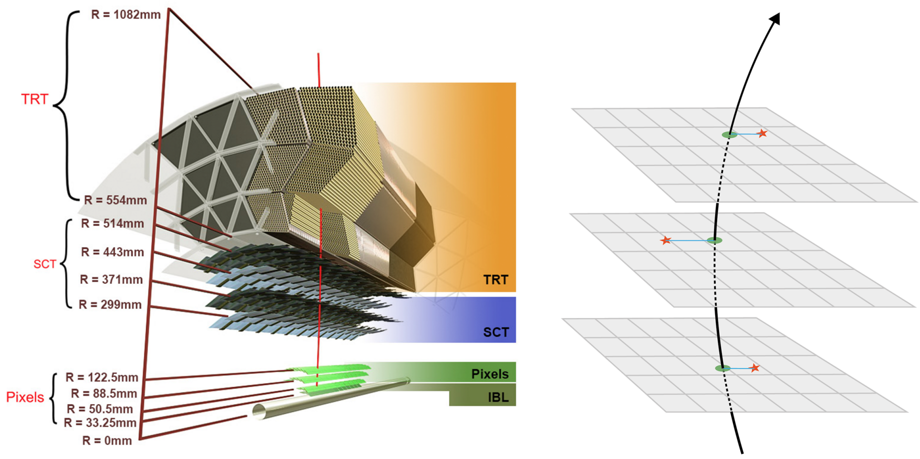


Figure 3.1.: Schematic cross-section of the inner detector of the ATLAS detector (© CERN).

The inner detector (ID) is the innermost part of the ATLAS detector. It is used for tracking charged particles, particle identification, and primary and secondary vertex determination [50]. The ID consists of three different sub-detectors. Their rough structure is depicted in Figure 3.1. The ID is encapsulated with a solenoid magnet providing a 2 T axial magnetic field on the inside of the ID. The magnetic field is crucial to measure the momentum of a charged particle. The transverse momentum is measured as the curvature of the particle's track due to the Lorentz force.

The detector part closest to the beam pipe is the pixel detector. It consists of 1744 modules arranged in three barrel layers. Each module hosts 47232 silicon pixels with a size of $50 \times 400\,\mu\text{m}^2$. The pixels are semiconductor trackers used to detect the traversing of charged particles.

The following part of the detector is the semiconductor tracker (SCT). It consists of 4088 modules arranged in four layers to guarantee four position measurements of charged particles. Each module consists of four silicon sensors. The sensors offer a 17 µm resolution in-plane lateral and 580 µm in-plane longitudinal.

The outermost part of the inner detector is the transition radiation tracker (TRT). It consists of polyimide drift (straw) tubes with a 4 mm diameter that are arranged in a 528 mm thick cylindrical layer around the beam pipe. The straw tubes are interleaved with fibres for the readout. The transition radiation tracker utilises the transition light emitted by charged particles traversing the interface between two media with different indices of refraction. The TRT offers a measurement of charged particles and electron identification.

3.2.3. Calorimeter System

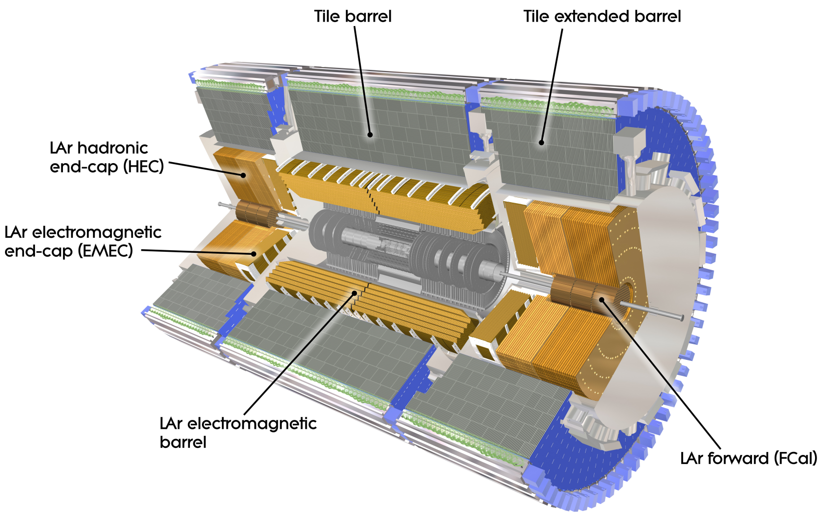


Figure 3.2.: Computer generated image of the ATLAS calorimeter (© CERN).

The calorimeters are the detector layers following the inner detectors. Its structure is divided into the Electromagnetic calorimeter (ECal) and the hadronic calorimeter (HCal).

Electromagnetic Calorimeter

The ECal is used for the energy and position measurement of electric charged particles and photons. It utilises bremsstrahlung and pair production to create a cascade of charged particles which are measured. It offers full azimuthal coverage and is equipped with end caps in the longitudinal direction of the beam pipe. The ATLAS ECal is a sampling calorimeter operating with lead as the passive and liquid Argon as the active medium. The innermost part of the ECal is a presampler which detects if the particle started showering before reaching the ECal.

Hadronic Calorimeter

The HCal measures the energy and position of baryons and mesons through strong interactions with the nuclei. In the range of $|\eta| < 1.7$ it is a sampling calorimeter with steel as the passive and scintillators as the active medium. For the end cap, liquid Argon is deployed as the active medium. The HCal works with the same principle as the ECal but offers less precise measurements.
In the range of $3.1 < |\eta| < 4.9$ ECal and HCal are substituted with the forward calorimeters (FCal) which are made up of three modules to fulfil the function of both calorimeters.

In combination with the FCal a total range of $|\eta| < 4.9$ is covered by the calorimeters. A computer-generated image of the structure of the different calorimeters used in the ATLAS detector is shown in Figure 3.2.

3.2.4. Muon Spectrometer

The muon spectrometer is the outermost part of the ATLAS detector. Its purpose is to detect charged particles exiting the calorimeters and measure their momentum in the range of $|\eta| < 2.7$. A detector dedicated to the measurement of muons is necessary because their mass makes them minimal ionizing particles in the energy range of the LHC collisions. The amount of energy loss due to bremsstrahlung is not sufficient to develop showers necessary to measure their energy. The muons' transverse momenta can be measured in the ID. Like the inner detector, the muon spectrometer utilises a magnetic field to conduct a measurement of the particle's momentum. In the muon spectrometer, however, a solenoid magnet is used to allow for the measurement of the muon's momentum along a different direction. Combining the measurements, one obtains full knowledge of the muons four-momentum. Further, does the high rate of stopped electrons and hadrons in the calorimeters enable a high specificity in the muon detection.

3.2.5. Trigger System

With a bunch spacing of 25 ns [48] collisions happen at a rate of 40 MHz. Each collision involves up to hundreds of particles. To reduce the data to a feasible amount, triggers are employed to filter less interesting events. Different layers of triggers operate either at the hardware or software level. The L1 trigger is a hardware level trigger and acts as the first filter for the events. It makes decisions in less than 2.5 µs. The subsequent L2 trigger is a software level trigger and makes decisions in less than 200 µs. Combined, the trigger system reduces the event rate from 40 MHz to 1000 Hz which are then stored at the CERN data centre.

4. Machine Learning in Particle Physics

Since this work focuses on improving the event reconstruction of dileptonic $t\bar{t}$ decays using machine learning techniques, this chapter aims to provide an overview over the fundamental concepts and methodologies employed in machine learning.
From a computational perspective, machine learning can be viewed as the optimization of a function $f : \mathbb{R}^n \to \mathbb{R}^m$ that maps input data points $\mathbf{x} \in \mathbb{R}^n$ to output predictions $\mathbf{y} \in \mathbb{R}^m$. The function f is parameterised by a set of learnable parameters θ, which are adjusted during the training process to minimise a predefined *loss function* $L(\mathbf{y}, f(\mathbf{x}; \theta))$. The loss function quantifies the discrepancy between the predicted outputs and the true target values, guiding the optimization process.

4.1. Supervised Learning

Machine learning can be broadly categorised into supervised and unsupervised learning. In supervised learning, the model is trained on a labelled dataset, where each input data point is associated with a corresponding target output. The goal of the model is to learn a mapping from inputs to outputs, enabling it to make accurate predictions on unseen data.

4.1.1. Monte Carlo Training Data

Events are usually simulated using a cascade of different programs, each simulating a different state of the event. First the hard scattering process is simulated using matrix element generators. These programs calculate the probabilities of different particle interactions based on the underlying physics theories, such as the Standard Model. Using these probabilities, they generate events that represent the initial state of the particles after the collision. The possible kinematic phase space is sampled according to these probabilities, resulting in a set of particles with specific momenta and energies. This type of event variables is often referred to as *parton-level*.
Next, the parton showering and hadronization processes are simulated. Parton showering models the emission of additional particles from the initial partons, while hadronization simulates the formation of hadrons from quarks and gluons. These processes are crucial for accurately modelling the final state particles observed in detectors.
Finally, detector simulation programs are used to simulate the interaction of particles with the detector material, producing realistic detector responses. This includes simulating the energy deposits in calorimeters, hits in tracking detectors, and other relevant signals.

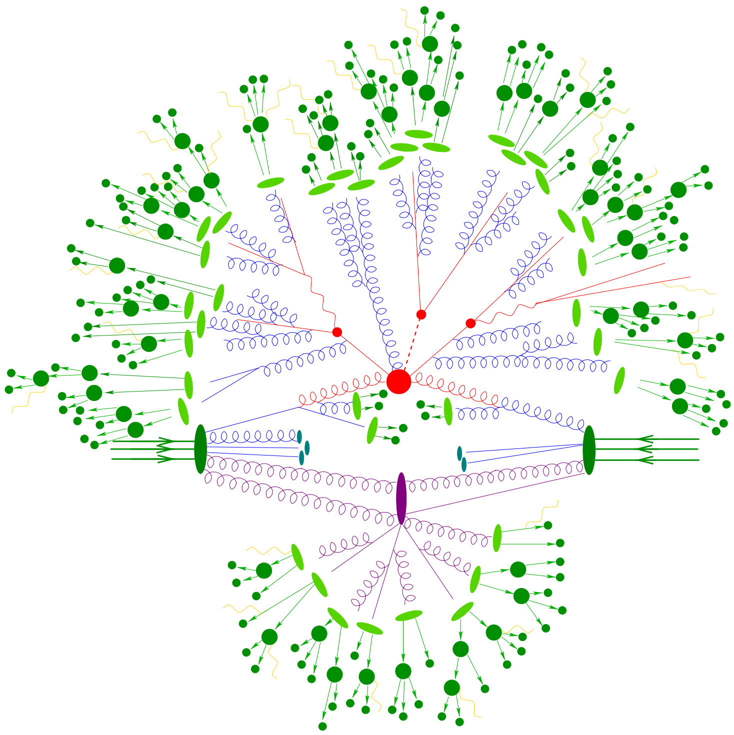


Figure 4.1.: Schematic representation of the Monte Carlo event generation process in particle physics. Colouring of vertices and particles indicates the event generation stage: hard scattering process (big red blob), QCD radiation (blue), hadronisation (light green), showering (dark green), and a secondary vertex (purple). The Figure is taken from Reference [51].

While the actual detector response is simulated using complex detector simulation software, for many machine learning applications, a simplified representation of the detector response is sufficient. This can involve smearing the particle momenta and energies according to the detector resolution, applying efficiency corrections, and simulating the effects of pile-up.
This is because, for the reconstruction of the physics objects, such as jets, leptons, and missing transverse energy, highly sophisticated algorithms are used that already take into account the detector effects (Note that these algorithms may also be machine learning based). Therefore, the input features for machine learning models can often be derived directly from these reconstructed objects, rather than relying on the raw detector signals. The reconstructed object event variables are called *reco-level*.

4.1.2. Training

During the training phase, the model is presented with a set of input features derived from the reco-level event variables, along with their corresponding target outputs, which are typically derived from the parton-level event variables. The model learns to map the input features to the target outputs by minimizing a loss function that quantifies the difference between the predicted and true values. Common loss functions include mean squared error for regression tasks and cross-entropy loss [52] for classification tasks. The training process involves iteratively (each step is called *epoch*) updating the model's parameters using an optimization algorithm to minimise the loss function.

4.1.3. Validation and Testing

To evaluate the performance of the trained model, it is essential to validate and test it on independent datasets that were not used during training. This helps to assess the model's generalization capabilities and ensures that it can make accurate predictions on unseen data. A common practice is to split the available data into training, validation, and test sets. The validation set is used to tune hyperparameters and monitor the model's performance during training, while the test set is reserved for the final evaluation of the model's performance.
Since in particle physics, simulated data is often used for training and background modelling, one typically employs a k-folding strategy [53], where the data is divided into k subsets. The model is trained k times, each time using a different subset as the validation set and the remaining subsets for training. This approach helps to use the available data more efficiently.

4.2. Regression and Conditional Density Learning

To highlight some important distinctions between different machine learning tasks, it is useful to briefly discuss regression and conditional density learning.

4.2.1. Supervised Learning as Risk Minimisation

In supervised learning, the goal is to learn a parametrised function f_θ that maps input data $\mathbf{x}$ to output predictions $\mathbf{y}$. This can be formalised as a risk minimisation problem, where the objective is to minimise the expected loss over the joint distribution of inputs and outputs. The expected risk can be expressed as:

$$R(f) = \mathbb{E}_{\mathbf{x},\mathbf{y}}[L(\mathbf{y}, f_\theta(\mathbf{x}))] \tag{4.1}$$

where L is the loss function that quantifies the discrepancy between the true output $\mathbf{y}$ and the predicted output $f_\theta(\mathbf{x})$. The goal of training is to find the parameters θ that minimise this expected risk, which is typically approximated using empirical risk minimisation on the training dataset.

4.2.2. Regression

In regression tasks, the output $\mathbf{y}$ is a continuous variable, and the model is trained to predict a specific value for each input $\mathbf{x}$. The loss function commonly used for regression is the mean squared error (MSE), which measures the average squared difference between the predicted values and the true values:

$$L_{\mathrm{MSE}}(\mathbf{y}, f(\mathbf{x})) = \frac{1}{N}\sum_{i=1}^{N}(y_i - f_\theta(\mathbf{x}_i))^2 \tag{4.2}$$

where N is the number of samples in the dataset. The model learns to minimise this loss function. If the labels are not deterministic, meaning that there is inherent uncertainty in the target values, one can describe the labels using a conditional probability distribution $p(\mathbf{y}|\mathbf{x})$. In this case, minimising the MSE corresponds to learning the conditional mean of the distribution (see Chapter B for a proof):

$$f_\theta(\mathbf{x}) = \mathbb{E}[\mathbf{y}|\mathbf{x}] = \int \mathbf{y} p(\mathbf{y}|\mathbf{x}) d\mathbf{y}\,. \tag{4.3}$$

Depending on the loss-function used, the model can learn different statistics of the conditional distribution. For example, using the mean absolute error (MAE) loss function would lead to learning the conditional median instead of the mean.

4.2.3. Conditional Density Learning

In contrast to regression, conditional density learning aims to learn the entire conditional probability distribution $p(\mathbf{y}|\mathbf{x})$ rather than just a point estimate. This is particularly useful when the target variable has inherent uncertainty or when multiple outcomes are possible for a given input. To quantify the difference between the true conditional distribution and the model's predicted distribution, the Kullback-Leibler (KL) divergence is often used as a loss function [54]

$$L_{\mathrm{KL}}(p, q) = \int p(\mathbf{y}|\mathbf{x}) \log\left(\frac{p(\mathbf{y}|\mathbf{x})}{q_\theta(\mathbf{y}|\mathbf{x})}\right) d\mathbf{y} \tag{4.4}$$

where $p(\mathbf{y}|\mathbf{x})$ is the true conditional distribution and $q_\theta(\mathbf{y}|\mathbf{x})$ is the model's predicted distribution. Minimizing this loss function encourages the model to produce a predicted distribution that closely matches the true distribution, allowing for a more comprehensive understanding of the uncertainty and variability in the target variable. In practice, computing KL divergence can be challenging, since the true distribution might be unknown or difficult to estimate. In such cases, one relies on the data to provide an empirical estimate of the true distribution, and the loss function is approximated using samples from the dataset. It can be shown that minimizing the KL divergence is equivalent to maximizing the likelihood of the observed data under the model's predicted distribution, which is a common approach in training probabilistic models. For the case of a discrete target variable, this corresponds to minimising the cross-entropy loss, which is widely used in classification tasks [52].

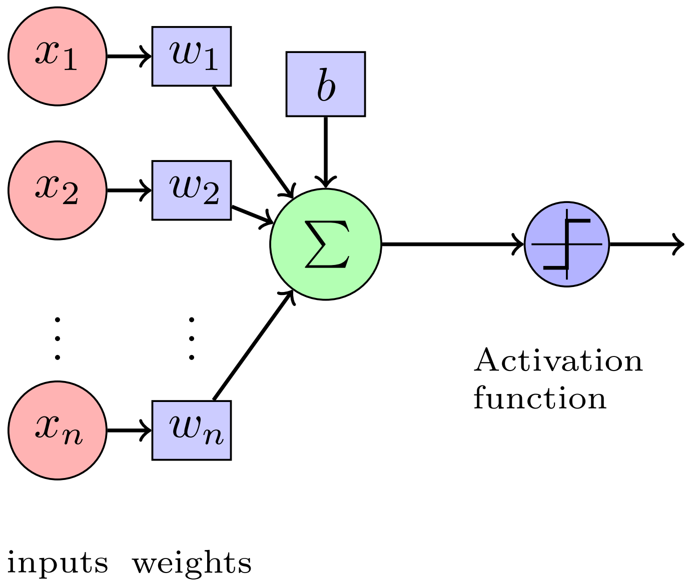


Figure 4.2.: Schematic of a single artificial neuron (perceptron). The neuron receives multiple input signals, each associated with a weight, and computes a weighted sum of these inputs, optionally a so-called bias is added. An activation function is then applied to this sum to produce the neuron's output.

4.3. Neural Networks

As elaborated before, machine learning models can be viewed as parameterised functions that map input data to output predictions. Neural networks are a class of machine learning models that are inspired by the structure and function of biological neural networks in the human brain. They consist of interconnected artificial neurons that process and transform the input data to produce the desired output. The simplest unit of a neural network is the artificial neuron, also known as a perceptron [55]. A perceptron takes multiple input signals, each associated with a weight, and computes a weighted sum of these inputs. An activation function is then applied to this sum to produce the neuron's output. This structure is illustrated in Figure 4.2. The general structure of a perceptron is inspired by biological neurons. The activation function introduces non-linearity into the model, allowing it to learn complex patterns in the data. Common activation functions include the sigmoid function, hyperbolic tangent (tanh), and rectified linear unit (ReLU) [56]. For this work, the ReLU activation function is used, which is defined as $f(x) = \max(0, x)$, meaning that it outputs the input directly if it is positive, and zero otherwise. This activation function has been shown to be effective in training deep neural networks by mitigating the vanishing gradient problem [56].

4.3.1. Dense Neural Network

A dense neural network (DNN), also known as a fully connected neural network [57], is a type of artificial neural network where each neuron in one layer is connected to every

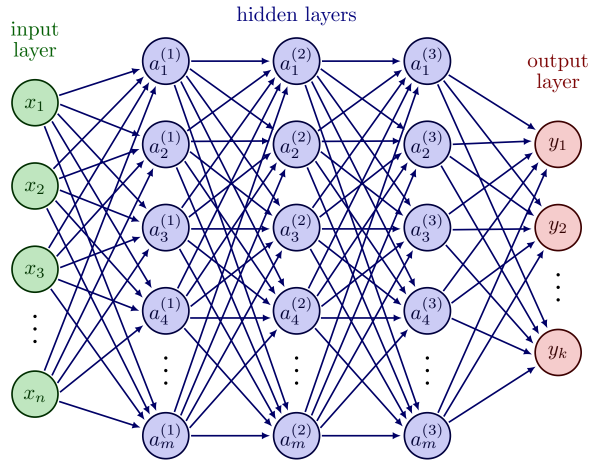


Figure 4.3.: Schematic of a dense neural network with multiple layers. Each layer consists of multiple neurons, and each neuron in one layer is connected to every neuron in the subsequent layer.

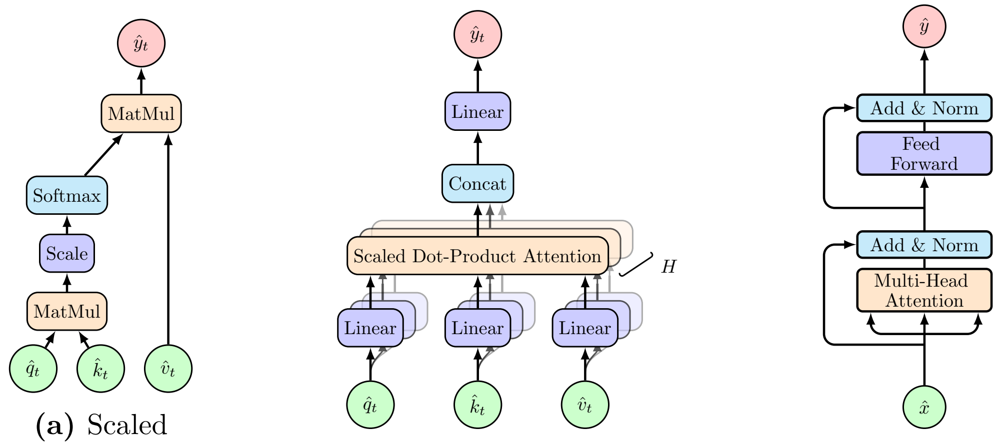
neuron in the subsequent layer. This architecture allows for the learning of complex relationships between input features and output targets. A schematic of a dense neural network is shown in Figure 4.3.

A DNN consists of an input layer, one or more hidden layers, and an output layer. The input layer receives the input features, while the hidden layers perform a series of transformations on the data using the weights and biases associated with each neuron. The output layer produces the final predictions. Each layer applies an activation function to the weighted sum of inputs from the previous layer, enabling the network to learn non-linear mappings. The depth (number of layers) and width (number of neurons per layer) of the network can be adjusted.

4.3.2. Transformer Networks

The transformer architecture deals with sequential data using an attention mechanism. This allows transformers to process information in parallel, making them efficient for training on large datasets. Additionally, transformer can capture long-range dependencies in the data very effectively, as they relate all elements of the input sequence to each other through attention mechanisms.

The core component of the transformer architecture is the attention mechanism, which allows the model to focus on different parts of the input sequence when making predictions. The most commonly used attention mechanism is the scaled dot-product attention [58], illustrated in Figure 4.4a. In this mechanism, three vectors are computed for each input element: the query (q), key (k), and value (v) vectors. The attention scores are calculated by taking the dot product of the query and key vectors, scaling them by the square root of the dimension of the key vectors, and applying a softmax function to obtain attention weights. These weights are then used to compute a weighted sum of the value vectors, producing the output of the attention mechanism. Multi-head attention, shown in Figure 4.4b, extends the scaled dot-product attention by allowing the model to attend



(a) Scaled Dot-Product Attention **(b)** Multi-Head Attention **(c)** Transformer Layer

Figure 4.4.: Different components of the transformer architecture from smallest to largest unit: (a) Scaled Dot-Product Attention, (b) Multi-Head Attention, and (c) Transformer Layer.

to different parts of the input sequence simultaneously. This is achieved by using multiple parallel attention heads, each with its own set of learned linear transformations for the query, key, and value vectors. The outputs of all attention heads are concatenated and linearly transformed to produce the final output.
The key, query and value vectors are typically obtained by applying learned linear transformations to the input embeddings. Depending on the type of relational information one wants to capture, one can choose to derive these vectors from different input sources. For example, in the *self-attention mechanism*, all three vectors are derived from the same input sequence ($\hat{q}_t = \hat{k}_t = \hat{v}_t = \hat{x}_t$), allowing the model to relate different elements within the same sequence. In *cross-attention*, the query vectors are derived from one sequence, while the key and value vectors are derived from another sequence ($\hat{q}_t = \hat{x}_{1,t}$, $\hat{k}_t = \hat{v}_t = \hat{x}_{2,t}$), enabling the model to relate information between two different sequences.

Symmetries In many applications, including particle physics, the input data exhibits certain symmetries that can be exploited to improve model performance.
The attention mechanism has inherent permutation equivariances, that can be exploited. Self-attention is permutation equivariant to the input sequence, meaning that permuting the input elements results in a corresponding permutation of the output elements. This property is particularly useful when dealing with sets or unordered data, where the order of the elements does not carry any inherent meaning. Cross-attention is permutation equivariant to permutations of the query inputs and separately to permutations of the key/value inputs. This allows the model to effectively relate information between two different sequences, regardless of their order.

4.3.3. Training Neural Networks

Training neural networks involves optimizing the model's parameters (weights and biases) to minimise the loss function. The number of parameters in a neural network can rise rapidly, especially in deep architectures, making the optimization process computationally intensive. The most commonly used optimization algorithm for training neural networks is stochastic gradient descent (SGD) [59], which iteratively updates the model's parameters based on the gradients of the loss function with respect to the parameters. Modern variants of SGD, such as Adam [60], incorporate adaptive learning rates and momentum to improve convergence. The backpropagation algorithm [57] is used to efficiently compute the gradients of the loss function with respect to the model's parameters. It involves propagating the error signal backward through the network, allowing for the calculation of gradients at each layer. These gradients are then used by the optimization algorithm to update the parameters. The algorithm relies on the chain rule and nested structure of the neural network to compute the gradients efficiently. Regularization techniques, such as dropout [61] and weight decay, are often employed during training to prevent overfitting and improve the model's generalization capabilities. Dropout randomly deactivates a fraction of the neurons during training, forcing the network to learn more robust features. Weight decay [62] adds a penalty term to the loss function based on the magnitude of the weights, discouraging overly complex models and promoting generalization.

5. Analysis and Reconstruction of Dileptonic Top-Quark Pair Events

5.1. Data and Simulation

This section describes the data and simulation samples used in this analysis. At the current stage of this thesis, only simulation samples have been studied; the data samples will be described in the final version. The simulated samples are produced using Monte Carlo (MC) event generators as described in Section 4.1.1.

While a complete analysis requires detailed investigation of background processes contributing to the signal regions, this thesis focuses on developing and evaluating reconstruction methods. Therefore, only the signal process of $t\bar{t}$ production with dileptonic decays is described here. The background processes and their simulation will be discussed in the final thesis.

5.1.1. Simulation of Dileptonic Top-Quark Pair Events

The signal process of $t\bar{t}$ production with dileptonic decays is simulated using Powheg [63] at next-to-leading order (NLO) in perturbative quantum chromodynamics (QCD). The Powheg generator is interfaced with Pythia [25] for parton showering, hadronization, and underlying event modelling.

To link the reco-level particles to the parton-level objects, a parton matching is performed. The parton-level objects are defined as the top quarks and their decay products after final state radiation before hadronization. The matching is done by iteratively associating reconstructed jets to partons based on angular distance criteria. The decay products of the top quarks are matched within a cone of $\Delta R \leq 0.3$ and $\Delta R \leq 0.1$ for jets and leptons (e,μ) respectively. If multiple reconstructed objects are found within this cone, the one with the smallest ΔR is chosen.

5.1.2. Simulation of Non-Relativistic Quasi-Bound State Resonance

5.1.3. Object Definition

The reconstruction of physics objects—electrons, muons, jets, and missing transverse energy ($\not{E}_{\mathrm{T}}$)—is performed using the standard Atlas reconstruction algorithms [50].

Electrons are required to have transverse momentum $p_{\mathrm{T}} > 5$ GeV and pseudorapidity $|\eta| < 2.47$, excluding the transition region between the barrel and endcap calorimeters

($1.37 < |\eta| < 1.52$). Electrons must satisfy the Tight identification criteria and be isolated from other activity in the detector.

Muons are reconstructed using information from both the inner detector and the muon spectrometer. They are required to have $p_{\mathrm{T}} > 5$ GeV and $|\eta| < 2.5$. Muons must satisfy the Tight identification criteria and be isolated.

Jets are reconstructed using the anti-k_t algorithm [64] with a radius parameter of $R = 0.4$. They are required to have $p_{\mathrm{T}} > 25$ GeV and $|\eta| < 2.5$. b-tagging is performed using the GN2 algorithm [65] to identify jets originating from b-quarks.

Missing Transverse Energy ($\not{E}_{\mathrm{T}}$) is calculated from the negative vector sum of the transverse momenta of all reconstructed objects in the event, including a soft term to account for energy not associated with reconstructed objects.

5.1.4. Event Selection

Events are selected to contain exactly two oppositely charged leptons (electrons or muons). One of the leptons must have $p_{\mathrm{T}} > 25$ GeV, while the other must have $p_{\mathrm{T}} > 5$ GeV. At least two jets are required, with at least one being b-tagged at the 77% working point (WP). Additional selection criteria are applied to suppress background contributions, which will be detailed in the final thesis.
For the training and evaluation of the reconstruction methods, only events that pass the selection criteria and have a successful parton matching are considered. Fully parton-matched events, make up about 60% of the whole sample. Events with jets or leptons of the $t\bar{t}$-process end up outside the detector acceptance are typically the main cause for an event being unmatchable. Thus, training with only parton-matched events might introduce a bias with respect to certain event topologies, if they are disfavoured to have a parton-matching.

5.2. Reconstruction of Top-Quark Pairs

Reconstructing the kinematics of dileptonic $t\bar{t}$ events requires an assignment of reconstructed objects to the parton-level decay products. This section describes the different reconstruction methods developed and evaluated in this thesis.
While the leptons are directly identified from the reconstructed objects, the assignment of jets to the b-quarks from the top quark decays is ambiguous due to the presence of multiple jets in the event. Additionally, the two neutrinos from the W boson decays are not directly detected, leading to missing information in the event reconstruction. Based on this, the reconstruction breaks down into two main tasks:

Jet Assignment Assigning the reconstructed jets to the b-quarks from the top quark decays.

Neutrino Regression Estimating the momenta of the two neutrinos using the measured $\not{E}_{\mathrm{T}}$ and other event kinematics.

5.2.1. B-Tagging

The identification of b-jets is crucial for the reconstruction of $t\bar{t}$ events. Due to the long lifetime of b-hadrons, they travel a measurable distance from the primary vertex before decaying, leading to displaced vertices and tracks with large impact parameters. The GN2 algorithm [65] is used for b-tagging in this analysis, which combines various discriminating variables using a deep (graph) neural network to achieve high efficiency and purity in identifying b-jets. The performance of the b-tagging algorithm is characterised by its efficiency (the probability of correctly identifying a b-jet) and its misidentification rate (the probability of incorrectly tagging a non-b-jet as a b-jet).

5.2.2. Jet Assignment Methods

This thesis explores several methods for jet assignment, ranging from traditional algorithms to machine learning approaches. While the leptons can be assigned to the W bosons based on their charge, the assignment of jets to the b-quarks is not straightforward due to the presence of multiple jets in the event. All the methods presented here, utilise kinematic relationships to make an assignment which jet and lepton stem from the same top quark decay. From recent publications e.g. [1], two conventional methods have been implemented as baselines:

Jet-Selection Both methods start by selecting the two b-jet candidates in the event based on the b-tagging at the 77% WP. If more than two b-tagged jets are present, the two with the highest p_{T} are chosen. If only one b-tagged jet is found, the non-tagged jet with the highest b-tagging score is selected as the second candidate.

χ^2-Method

For both possible assignments of the two b-jet candidates to the parton-level b-quarks, the invariant masses of the reconstructed top quarks are computed,

$$\chi^2 = (m_{b_1,\nu,\ell^+} - m_t)^2 + (m_{b_2,\bar{\nu},\ell^-} - m_t)^2 \tag{5.1}$$

where $m_t = 172.5$ GeV is the top quark mass. The assignment with the smaller χ^2 value is chosen.

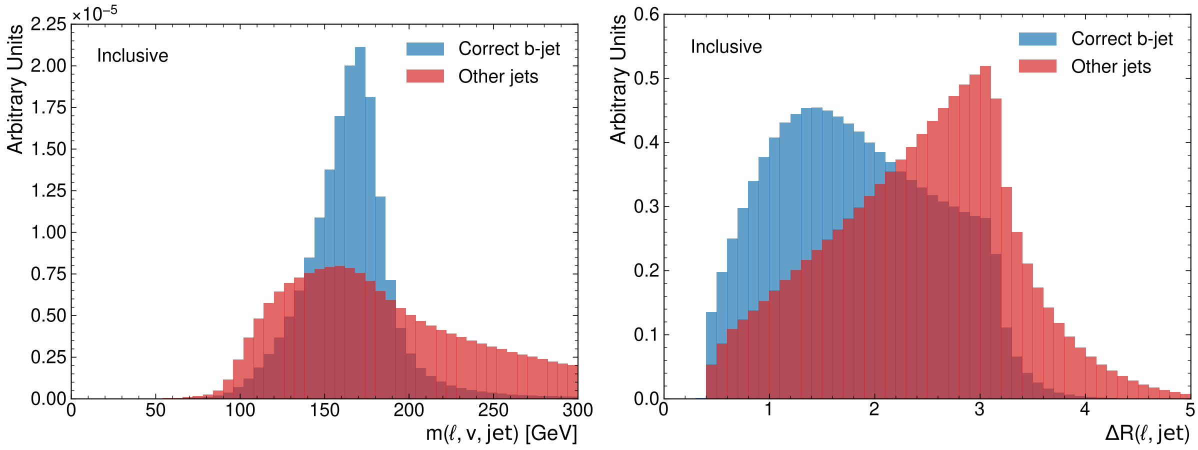
ΔR-Method

This method assigns the b-jet candidates based on their angular distance to the leptons.

$$\Delta R = \sqrt{(\Delta\phi)^2 + (\Delta\eta)^2} \tag{5.2}$$

The jet-lepton-pair with the smaller ΔR value is assigned to each other. The remaining jet is assigned to the other lepton.

Note, that the χ^2-Method relies on the reconstructed neutrino momenta to compute the invariant masses of the top quarks. Therefore, it is inherently linked to the neutrino regression method used. In contrast, the ΔR-Method is independent of the neutrino momenta. Further, it should be noted, that due to its reliance on the neutrino reconstruction



(a) χ^2-Method **(b)** ΔR-Method

Figure 5.1.: Distributions for pairing up correct and incorrect jet-lepton pairs for the reconstructed invariant mass of the top quark (a) and the ΔR between the jet and lepton (b). The correct pairs are shown in blue, while the incorrect pairs are shown in red. The distributions are normalised to unit area.

and the use of ν^2-Flows here, the χ^2-method should, technically, be considered a machine learning based approach too.
Empiric evidence for the validity of both can be found in Figure 5.1, which shows the distributions of the reconstructed invariant mass of the top quark and the ΔR between the jet and lepton for correct and incorrect jet-lepton pairs. The correct pairs show a clear peak around the top quark mass in the invariant mass distribution and a smaller ΔR compared to incorrect pairs, which supports the use of these variables for jet assignment. The ΔR-Method was used in previous ATLAS analyses [1] for events, that failed the kinematic reconstruction required for the χ^2-Method. However, in this thesis, both methods are evaluated on the full dataset for comparison.

5.2.3. Neutrino Regression Methods

Due to the limited availability of kinematic reconstruction methods in the ATLAS software framework, only one method for neutrino regression has been implemented and evaluated in this thesis. For completeness, two additional methods that have been used in previous ATLAS analyses are described here, but they have not been implemented or evaluated in this thesis.

Ellipse-Method

The ellipse method [43] is a kinematic reconstruction technique that exploits the constraints of the event to solve for the neutrino momenta. The method uses the measured E_T^{miss} and the momenta of the leptons and jets to construct a system of equations based on the invariant mass constraints of the W bosons and top quarks. By solving this system of equations, one can obtain possible solutions for the neutrino momenta. The method typically yields multiple solutions, and additional criteria are used to select the most likely solution based on the event kinematics. In previous ATLAS analyses, the solution yielding the lowest $m(t\bar{t})$ was chosen as the reconstructed neutrino momenta.

Neutrino-Weighter

The Neutrino-Weighter, first described in [44], is a method that assigns weights to different possible neutrino momentum configurations based on how well they satisfy the kinematic constraints of the event. The method uses the measured $\not{E}_{\mathrm{T}}$ and the momenta of the leptons and jets to compute a weight for each possible neutrino momentum configuration. For each configuration, a weight is computed based on how well the reconstructed event kinematics match the expected kinematics of a $t\bar{t}$ event - this includes constraints such as the invariant mass of the W bosons and top quarks. The configuration with the highest weight is then selected as the reconstructed neutrino momenta. This method is computationally intensive, as it requires scanning over a large number of possible neutrino momentum configurations.

ν^2-Flows

The method ν^2-Flows [45] uses normalizing flows to model the conditional probability distribution of the neutrino momenta given the observed event kinematics. The model is trained on simulated dileptonic $t\bar{t}$ events, where the true neutrino momenta are known from the parton-level information. The trained model can then be used to sample possible neutrino momenta for new events, allowing for a probabilistic reconstruction of the event kinematics. For each event, multiple samples of neutrino momenta are drawn from the model, and the one with the highest probability density is selected as the reconstructed neutrino momenta. A detailed outline of the ν^2-Flows method and its implementation can be found in Reference [45]. For this thesis, this method serves to establish a baseline for neutrino regression to evaluate the impact of improved jet assignment methods. Note, however, that this method has not been used in ATLAS analyses yet. However, due to matters of software compatibility, it was the only viable option at this stage of the thesis. Since ν^2-Flows processes the full event information to make a prediction for the neutrino momenta, it does not require a jet-assignment. Therefore, it can be used in conjunction with both the χ^2-Method and the ΔR-Method for jet assignment, allowing for a direct comparison of the impact of the jet assignment on the overall reconstruction performance.

6. ML-Based Reconstruction of Dileptonic Top-Quark Pair Events

In addition to the outline reconstruction methods described in Chapter 5, this chapter focuses on the development of a machine learning-based reconstruction strategies for the dileptonic $t\bar{t}$ channel. Machine learning-based reconstruction techniques have gained significant traction in recent years. One major example for this, is their use in flavour tagging algorithms [66]. For this, machine learning models are trained to address both challenges of jet assignment and neutrino regression, by learning from simulated data to predict the correct jet-parton assignments and neutrino momenta.
For the high permutation multiplicity of jet assignments in all-hadronic channels, SPANet [6] has demonstrated significant improvements over traditional methods. These architectures leverage attention mechanisms to improve the accuracy by leveraging the full event information and utilising permutation equivariant architectures to handle the combinatorial nature of the problem. Based on these successes, this thesis explores the adaptation of similar architectures for the dileptonic $t\bar{t}$ channel, which presents unique challenges due to the presence of two neutrinos and fewer jets. This topic has been largely understudied in the literature, with most efforts focusing on neutrino regression. Thus, this work aims to fill this gap by developing a machine learning model specifically tailored for the jet assignment problem in the dileptonic channel, and investigating how it can be combined with neutrino regression methods to offer a full reconstruction pipeline.
On the other hand, for neutrino regression, ν^2-Flows [45] has shown promise in improving the reconstruction of neutrino momenta by modelling the underlying probability distributions of the neutrino kinematics.
Despite both tasks being highly interconnected, both of them are mostly studied independently and the potential synergies between them remain largely unexplored. This thesis investigates how the jet assignment and neutrino regression tasks can be combined in a unified machine learning framework, allowing for a more holistic approach to event reconstruction in the dileptonic $t\bar{t}$ channel.

6.1. Jet-Assignment

Due to the wide range of available neutrino regression methods, the jet assignment problem while be studied independently first, before investigating how it can be combined with neutrino regression methods in a unified framework. In general, the task the model is trained to solve is to associate jets and lepton originating from the same top-quark decay. Different layouts to encode this task have been explored. After investigation, using a matrix

$$P_{pred}(\ell^{\pm}, j_i) \in [0,1], \quad i \in \{1, ..., N_{jets}\} \tag{6.1}$$

with the property

$$\sum_i P_{pred}(\ell^{\pm}, j_i) = 1\,, \tag{6.2}$$

as the networks output was chosen. This ensures, that each lepton is associated with exactly one jet. Since for the model training and evaluation only fully truth-matched events were used, the labels fulfil

$$P_{true}(\ell^{\pm}, j_i) \in \{0,1\}, \quad i \in \{1, ..., N_{jets}\} \tag{6.3}$$

with

$$\exists! j : P_{true}(\ell^{\pm}, j) = 1\,. \tag{6.4}$$

Thus, the problem becomes a multi-class classification for each of the two leptons. The labels are the one-hot encoded indices of the true matched jets for each lepton. For practical reasons, the labels are padded to all have the same length N_{max}. Events where the truth-match jet index is higher than the chosen padding-length were discarded for training and evaluation. This ensures, Equation (6.4) is fulfilled for all events.
During the process of devising the model's architecture, different neural network architectures were explored. To deal with the sequential nature of the jets, mostly recurrent and transformer networks were tested. Generally jet-permutation equivariant transformer networks proved to be most performant.

6.1.1. Neural Network Architectures

In Reference [6] it was investigated how permutation equivariance of transformer networks can greatly improve the performance and efficiency of the jet assignment in top-quark pair events. The SPANet-architecture was devised with the prospect of handling the high number of equivalent permutation of the jet assignment, which all lead to the same reconstructed tops. While the number of equivalent permutations was at least 20 in the all-hadronic decay channel, it is only 2 for the dileptonic decay channel in the lowest jet-multiplicity. Even though, it raises quadratically with the number of jets, this is reduced by using the b-tagging information available. Since this work investigates pairing up the jets and leptons, the only permutation equivariances are the jet-ordering and potentially the ordering of the leptons. The latter arises from the fact, that to a good approximation there is no difference between the decay properties of the top and anti-top respectively.
To investigate this, three different architectures for the transformer network were considered and evaluated.

FeatureConcatTransformer (FCT)

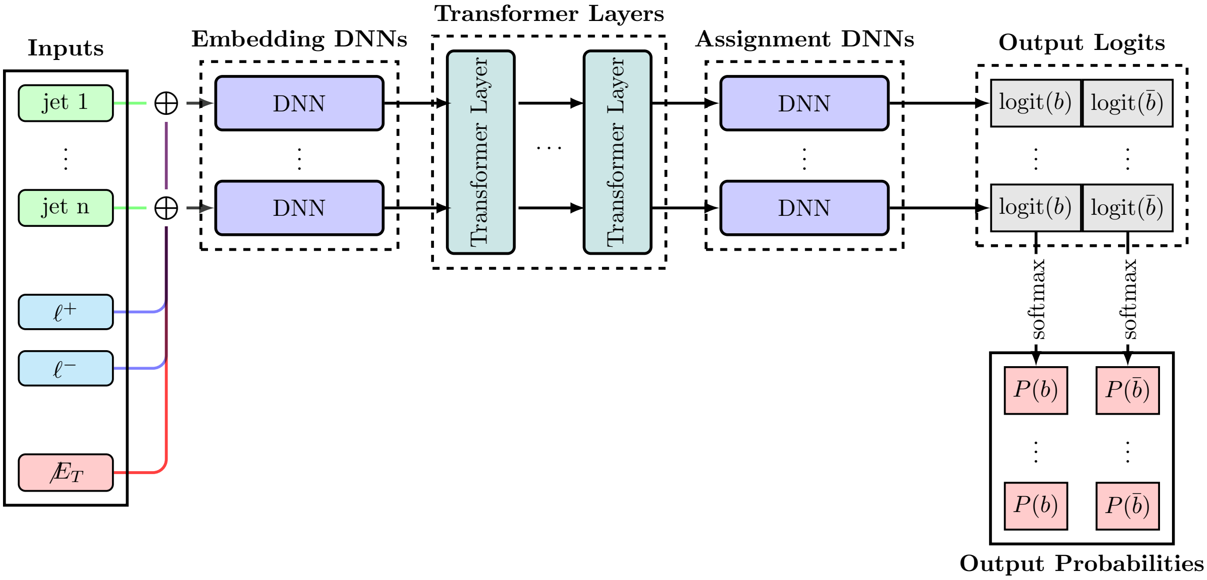


Figure 6.1.: This schematic visualises the general structure of the FeatureConcatTransformer architecture. A detailed description of the individual units making up this architecture is provided in this section. Note, that $\oplus$ represents a concatenation of the corresponding features.

The FeatureConcatTransformer (FCT) architecture is outlined in Figure 6.1. Its name-bearing component is the concatenation of the two lepton and the met features to each jet. This gives rise to a sequence of objects resembling the jets present in the event. By passing it through a shared DNN, this sequence is embedded to a fixed *Embedding Dimension* (ED), which is considered a hyperparameter of the model. The embedded sequence is passed through a number of transformer layers using self-attention [58]; the number of layers being another hyperparameter. The output elements of the central-transformer are lastly processed by another shared DNN, which provide two assignment scores for each jet. By computing a softmax function for each of the scores over all jets. Due to the properties of the softmax function, this provides a normalised score for each jet to be identified with the positive and negative lepton and bottom or anti-bottom respectively. The leptons are feed into the model with a fixed-ordering based on their charged. The model inherits its permutation equivariance with respect to the jet-ordering from the self-attention mechanism. Regarding the lepton ordering, the model is not permutation equivariant. Since the underlying physics presumably is, it can be expected that the network picks this structure up from the training data.

CrossAttentionTransformer (CAT)

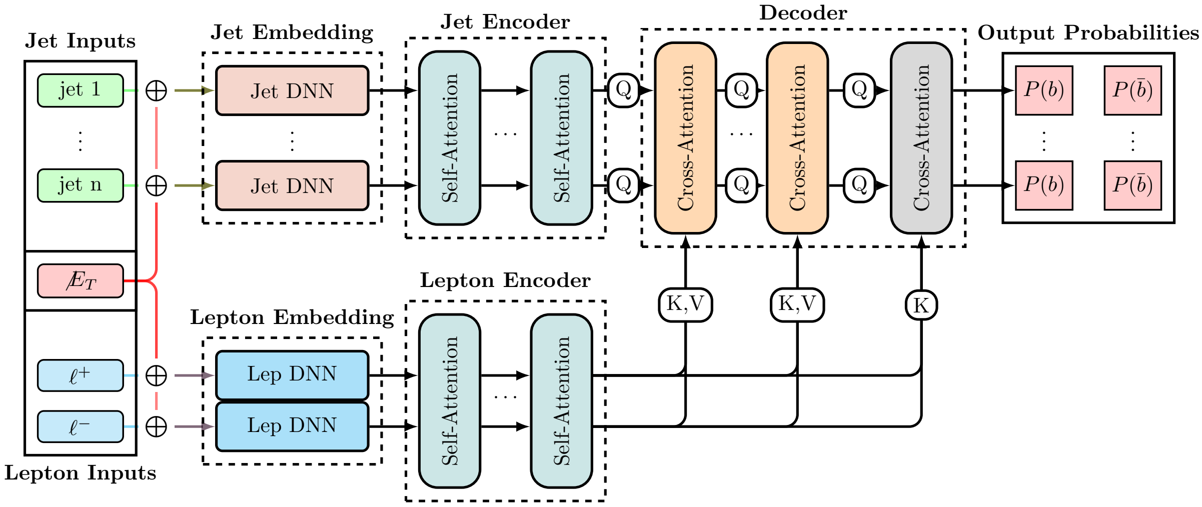


Figure 6.2.: This schematic visualises the general structure of the CrossAttentionTransformer architecture. A detailed description of the individual units making up this architecture is provided in this section. Note, that $\oplus$ represents a concatenation of the corresponding features.

The CrossAttentionTransformer (CAT) architecture is outlined in Figure 6.2. It is based on the idea of using cross-attention to directly model the interactions between the jets and leptons. Instead of concatenating the features, the leptons are treated as a sequence too. Instead, the met features are concatenated to both the jets and the leptons. This architecture then encodes the jets and leptons separately, using shared DNNs to embed them into a fixed *Embedding Dimension* (ED). The jets and leptons are then passed through separate transformer layers using self-attention, allowing the model to capture the internal relationships within each set of objects. The outputs of the jet and lepton transformers are then combined in a decoder using cross-attention layers, which allow the model to learn the interactions between the jets and leptons. Finally, the last layer outputs the cross-attention scores for each jet-lepton pair using dot-product attention followed by a softmax function to provide the final probabilities for the jet-lepton assignments. For simplicity, the number of encoder and decoder layers are chosen to be the same. In addition to the FCT's permutation equivariance with respect to the jet-ordering, the CAT architecture is also permutation equivariant with respect to the lepton-ordering, due to the separate encoding and self-attention layers for the leptons. This allows the model to fully capture the underlying symmetries of the problem, which can potentially lead to improved performance.

CompactTransformer (CPT)

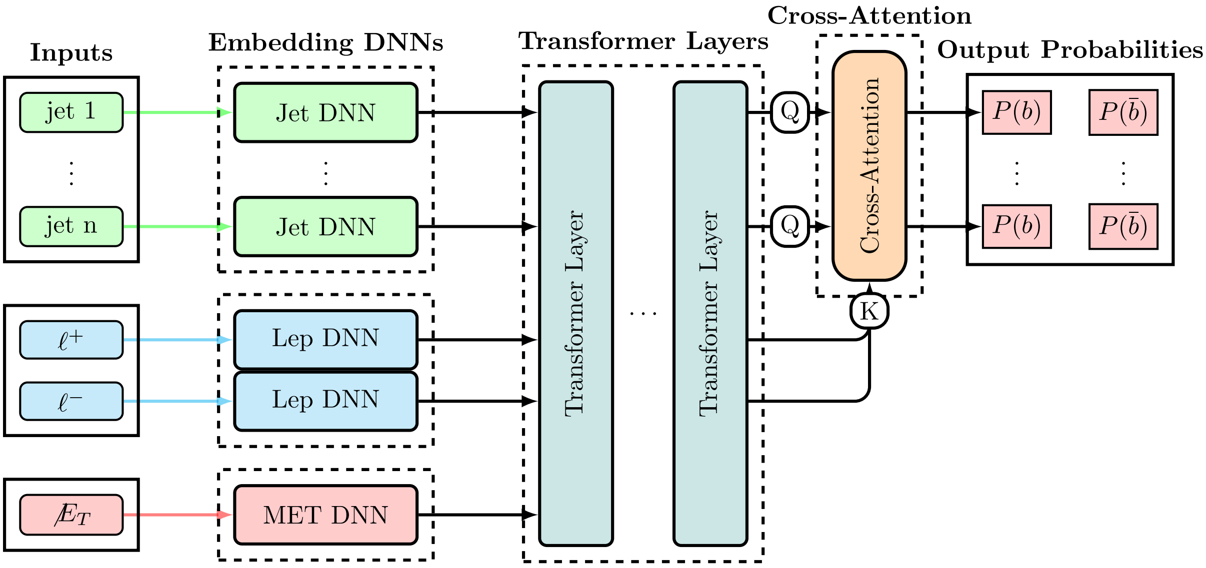


Figure 6.3.: This schematic visualises the general structure of the CompactTransformer architecture. A detailed description of the individual units making up this architecture is provided in this section.

The CompactTransformer (CPT) architecture is outlined in Figure 6.3. It is designed to be a sleeker version of the CrossAttentionTransformer (CAT) architecture, while still maintaining the permutation equivariance with respect to both the jet and lepton ordering. The CPT embeds jets, leptons, and MET features separately using shared DNNs, similar to the CAT architecture. However, instead of using separate transformer layers for the jets and leptons, the CPT combines them into a single sequence and passes it through a shared transformer layer using self-attention. This allows the model to capture both the internal relationships within each set of objects and the interactions between them in a more compact architecture. The final output layer provides the jet-lepton assignment scores using dot-product attention followed by a softmax function, similar to the CAT architecture. While the self-attention itself is permutation equivariant to the full sequence, the separate embeddings for the objects break this into permutation equivariance for the objects sharing the same embeddings—i.e., jets and leptons respectively.

Common Properties

For all models, the input features for the jets and leptons include their kinematic properties such as transverse momentum (p_T), pseudorapidity (η), azimuthal angle (ϕ), and energy (E). Additionally, the b-tagging for the jets is included as the continuous quantile of the corresponding b-tagging working point, and for the leptons, the charge is provided. The missing transverse energy (MET) features, including its magnitude and azimuthal angle, are also included as input to the model. Further, global event features such as the number of jets, the number of b-tagged jets, and the angular separation between the two leptons are included to provide additional context for the model.

Each model includes a layer dedicated to normalising the input features, which helps to improve the training stability and convergence. These are set once on the training data and are not updated during the training process. The normalisation is performed by computing the mean and standard deviation of each feature across the training dataset, and then using these statistics to standardise the features by subtracting the mean and dividing by the standard deviation. Further, quantities such as the p_T and energy of the jets and leptons are transformed using a logarithmic scale to better handle the wide range of values.
Using the model's forward pass, the jet-lepton assignment scores are computed for each jet and lepton pair. Next, each possible pairing of jets and leptons is scored by multiplying the probabilities for the positive and negative lepton. The pairing with the highest score is then selected as the model's prediction for the jet-lepton assignment. Solutions where the same jet is assigned to both leptons are discarded, as they are not physically meaningful. The maximum number of jets considered by the model is set to $N_{max} = 6$. This means that the model can handle events with up to 6 jets, which covers around 99.4% of dileptonic $t\bar{t}$ events.

6.1.2. Training Procedure

The training procedure for all three architectures is identical, as no major difference in the training behaviour was observed. The models are trained using a cross-entropy loss function for each lepton, which measures the discrepancy between the predicted probabilities and the true labels. The total loss is computed as the sum of the losses for both leptons and the loss per-event is scaled using sample weights. The sample weights are computed from combining the event weights with the inverse of the number of events per jet-multiplicity, which helps to mitigate potential biases in the training data and ensures that the model learns effectively from all events, regardless of their jet multiplicity. Updating of the model parameters is performed using the AdamW optimizer [62], which is a variant of the Adam optimizer that incorporates weight decay for regularisation. Further, a dropout with a rate of $r = 0.2$ is applied to all internal forward passes. The training is conducted over 100 epochs, with early stopping based on the validation loss to prevent overfitting. Additionally, learning rate scheduling is employed to adjust the learning rate during training, which can help improve convergence and overall performance. For the initial learning rate, a value of 10^{-4} is chosen, and the learning rate is reduced by a factor of 0.5 if the validation loss does not improve for 5 epochs. If the validation loss does not improve after 10 epochs, so 2 reduction of the learning rate, the training is terminated. The batch size is set to 4028, and the models are trained on a GPU to accelerate the training process.
During the training process, the model's performance is monitored on a separate validation set, which is composed of 20% of the initial training data, to ensure that it is not overfitting to the training data. The best-performing model is selected based on the lowest validation loss, and its performance is further evaluated on a separate test set to assess its generalization capabilities. Apart from the loss, other metrics such as the accuracy of the jet-lepton assignment are reported during the training process to provide additional insights into the model's performance. If not reported otherwise, each model is trained on

around 5M fully parton-matched dileptonic top-quark pair events taken from the nominal $t\bar{t}$ sample described in Section 5.1.1.

6.1.3. Hyperparameter Optimisation

To evaluate the performance of the different architectures, a hyperparameter optimisation is performed for each of them. The hyperparameters considered are the embedding dimension (ED) and the number of transformer layers. Other relevant hyperparameters, such as the learning rate, dropout rate, and batch size, were kept fixed across all models to ensure a fair comparison. The hyperparameter optimisation is conducted using a grid search approach, where a predefined set of values for each hyperparameter is evaluated. For all architectures, the embedding dimension is varied in the range of $\{16, 32, 64, 128\}$ and the number of transformer layers is varied in the range of $\{4, 6, 8, 10\}$. For the CAT architecture, the number of encoder and decoder layers are chosen to be equal. Each model and hyperparameter combination, the model is trained and evaluated 5 times with different random seeds and k-folded training/validation data to ensure the robustness of the results. The performance of the models is then compared based on the average validation loss and the accuracy of the jet-lepton assignment on a separate test set. From the variance of the results across different random seeds and data splits, the statistical significance of the differences in performance between the architectures can be assessed. This study allows for the identification of the most effective architecture and hyperparameter configuration, as well as insights into their scaling behaviour and robustness to different training conditions.

The individual results of the grid-search for each architecture are shown in Figures A.8 to A.10 in the Appendix Chapter A. From these results, it can be observed that the performance of all architectures generally improves with increasing embedding dimension. For the number of layers, the performance appears to be more stable across different values, with a slight improvement observed for higher numbers of layers. However, from some instances, it can be observed that the performance can degrade for very high numbers of layers, which may be due to overfitting or difficulties in training deeper models. Overall, the results suggest that while both hyperparameters have an impact on the model's performance, the embedding dimension appears to have a more significant effect compared to the number of layers.

Figure 6.4 compares the different architectures in terms of their (averaged) assignment accuracy as a function of the number of trainable parameters. Different aspects can be observed from these results. Regarding the individual architectures, it can be observed that the CPT architecture appears to be unstable for low embedding dimensions. While the other models always show very consistent results across different random seeds and data splits, the CPT architecture shows a large variance in performance for low embedding dimensions, which is reduced for higher embedding dimensions. This suggests that the CPT architecture may require a certain level of model complexity to perform well, and that it may be more sensitive to the choice of hyperparameters compared to the other architectures. For the FCT and CPT architectures, the performance appears to be more stable across different hyperparameter configurations, with a clear trend of improving performance with increasing number of model parameters and very small variation across

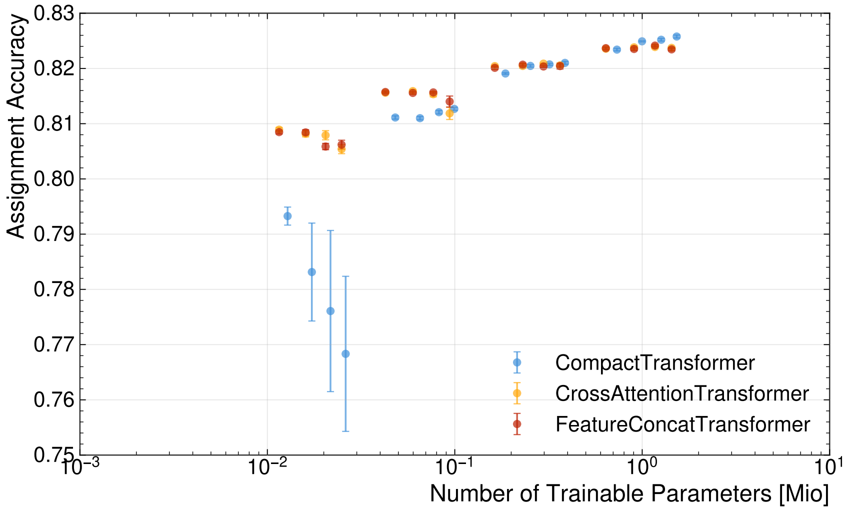


Figure 6.4.: Assignment Accuracy of all architectures as a function of the number of trainable parameters. Each point represents the average accuracy over 5 runs with different random seeds and k-folded training/validation data. The error bars indicate the standard deviation across the runs. Note, that the scaling of the x-axis is logarithmic, which allows for a better visualisation of the differences in performance across different model sizes.

training runs. This suggests that these architectures are more robust to the choice of hyperparameters and can achieve good performance even with smaller model sizes. Overall, it shows that the performance of the CPT architecture is more sensitive to the choice of hyperparameters, particularly the embedding dimension, compared to the FCT and CPT architectures. Partly, this can be explained by the fact, that the CPT architecture treat the jets, lepton and met features as one sequence, which may require a certain level of model complexity to effectively capture the interactions between these different types of features. In contrast, the FCT and CPT architectures treat the jets and leptons separately, which may allow them to learn more effectively from the data even with smaller model sizes.

This plot also reveals, why the performance varies more between different embedding dimensions than between different numbers of layers: the embedding dimension has a much larger impact on the number of trainable parameters compared to the number of layers, which can explain why it has a more significant effect on the model's performance. Note, that the number of parameters scales linearly with the number of layers, while it scales quadratically with the embedding dimension, which can explain the observed differences in performance across different hyperparameter configurations.

Apart from the instability of the CPT architecture for low embedding dimensions, the performance of the different architectures appears to be quite similar across different hyperparameter configurations. Generally, it emerges, that the performance seems to depend much more on the number of trainable parameters than on the specific architecture, with all architectures showing a similar performance for a given number of parameters. This suggests that the choice of architecture may be less important than ensuring that the model has sufficient capacity to learn the underlying patterns in the data, and that

the performance can be improved by increasing the model size rather than by changing the architecture. Also, the layout of the different models appears to be less important than the overall model capacity.
Figure 6.5 compares the different architectures in terms of their inference time as a func-

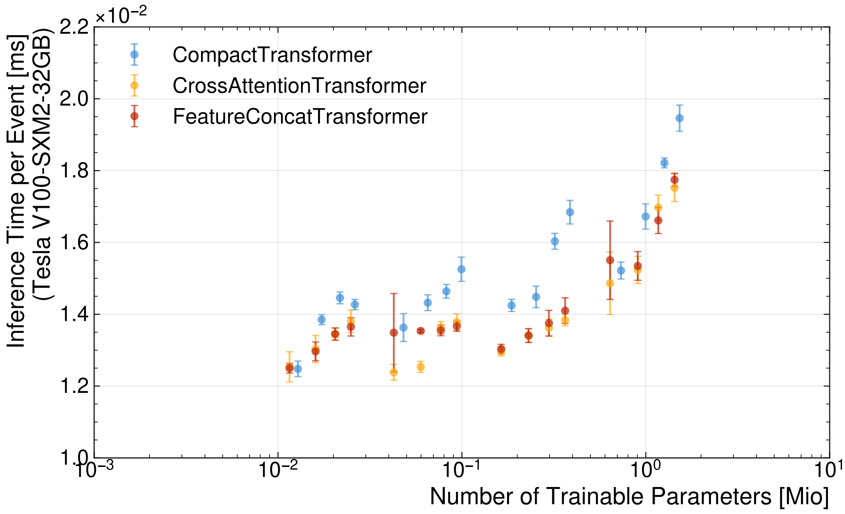


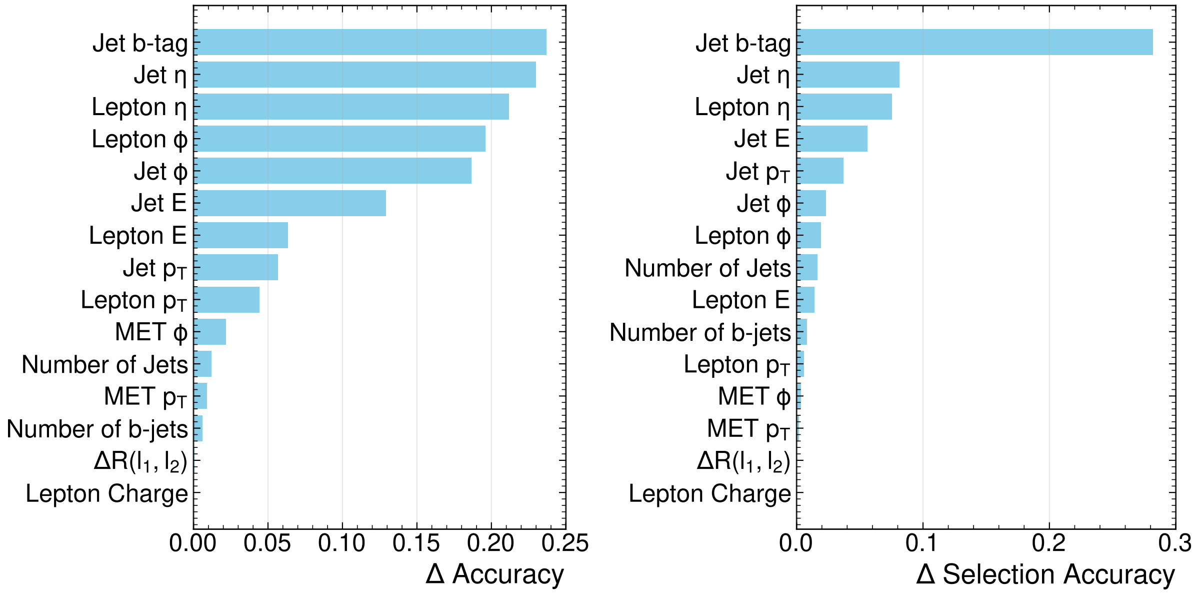
Figure 6.5.: Inference Time of all architectures as a function of the number of trainable parameters. Each point represents the average inference time over 5 runs with different random seeds and k-folded training/validation data. The error bars indicate the standard deviation across the runs. Note, that the scaling of the x-axis is logarithmic, which allows for a better visualisation of the differences in performance across different model sizes.

tion of the number of trainable parameters. The inference time is measured as the average time taken to process a single event during the evaluation phase, and it is an important metric for assessing the practical applicability of the models in real-time data processing scenarios. From the results, it can be observed that the inference time generally increases with the number of trainable parameters for all architectures, which is expected as larger models typically require more computational resources to make predictions. However, here, the number of layers shows to have a much more significant impact on the inference time compared to the embedding dimension, which can be explained by the fact that the number of layers directly affects the depth of the model and thus the number of computations required for each forward pass. In contrast, an increased embedding dimension has a less direct impact, as these computations can be parallelised more effectively, which can mitigate the increase in inference time. This is, however, only true for processing on highly parallelised hardware such as GPUs, and may not hold for processing on CPUs, where the increase in embedding dimension can lead to a more significant increase in inference time due to the limited parallelisation capabilities. Thus, these observed differences in inference time are to taken with a grain of salt, as they can be highly dependent on the specific hardware used for evaluation and may not generalise to other processing environments. Overall, it can be seen that the inference time raises most steeply for the CPT architecture. This can be explained by the fact that the CPT architecture combines

the jets, leptons, and MET features into a single sequence and passes it through a shared transformer layer using self-attention. The self-attention mechanism has a computational complexity that scales quadratically with the sequence length, which can lead to a significant increase in inference time as the number of features and model size increases. Thus, the slightly longer sequence length of the CPT architecture compared to the FCT and CAT architectures can contribute to its higher inference time, especially for larger model sizes. In contrast, the FCT and CAT architectures treat the jets and leptons separately, which can allow them to process the features more efficiently and reduce the overall inference time, even as the model size increases.
Combining the results from Figure 6.4 and Figure 6.5, it can be concluded that increasing the number of trainable parameters generally leads to improved performance in terms of assignment accuracy, but it also results in increased inference time. While the number of layers has a very limited effect on the assignment accuracy, it has a significant impact on the inference time. Thus, when selecting a model for practical applications, it is most often beneficial to choose a model with a higher embedding dimension and a lower number of layers, as this can provide a good balance between performance and computational efficiency. Additionally, the choice of architecture may also play a role in determining the optimal configuration, as some architectures may be more efficient than others in terms of inference time for a given number of parameters. To strike this balance, the CPT architecture with an embedding dimension of 128 and 6 layers is chosen as the best-performing model, as it achieves a high assignment accuracy while maintaining a reasonable inference time compared to the other configurations.

6.1.4. Feature Importance

Machine learning models are notoriously black-boxes, allowing very little insight into the underlying computation and decision-making. This is mostly, due to the abstract nature of the latent-spaces. However, over the years some techniques to gain insights into the model's computational structure have been developed. One prominent example of such techniques is measuring the *Feature Importance*. It aims to quantify how much each of the input features provided to the model impacts its computation and thus performance. The most robust framework for this, uses a game-theory approach using so-called SHAP-values [67, 68]. Due to the computational complexity of this method, for this thesis it was decided to make use of a simpler technique, called permutation importance [69]. This method quantifies this impact a feature has on the model's performance by measuring the difference in its performance upon effectively eliminating this feature from the inputs. The idea is to eliminate the feature by randomly shuffling its value among the events. Therefore, the feature's values become uncorrelated to the other features of the event and the model cannot make use of this feature. This was found to provide a robust way of accessing the impact of a feature. As the transformer model processes some features embedding in sequences, it was decided to permute the features among all the events and sequence elements. As the model is permutation invariant to jet-ordering, this aims to allow any correlations to be inherited by the jet-ordering. For this analysis, each of the input features was permuted and the difference in assignment and selection accuracy of the model compared to the baseline performance computed. This process



(a) Assignment Accuracy **(b)** Selection Accuracy

Figure 6.6.: The Plots (a) and (b) show the difference in assignment and selection accuracy, respectively, of the CPT-model when permuting the input features. For each feature, five different permutation are performed. The features are ranked in decreasing order of the observed reduction.

is repeated with five permutations to reduce stochastic variations from the individual permutations, and the difference is averaged over the trials. The results are presented in Figure 6.6, which shows the ranking and scores of the different features supplied to the model. Note, that this analysis only evaluates the impact of these variables in these trained models. In principle, it would be possible for a model trained without this variable to attain the same performance, by leveraging correlations of other variables. This is, why the aforementioned SHAP values compare combinations of variables and left out and often analyse, retraining a model without a certain feature. However, due to the large computational cost associated with this, such an investigation would exceed the scope of this thesis. For both metrics, the b-tag of the jets is the most important feature, followed by the jets' and leptons' pseudorapidity η respectively. While the order of the three leading features is the same for the two metrics, their individual impact is vastly different. For the assignment accuracy, all three features impose a similar drop in performance. For the selection accuracy, however, the difference in performance caused by disabling the b-tag is more than three times as large as the pseudorapidities. This highlights, that the selection accuracy is mainly carried by the b-tag, while for the assignment of the jets, their kinematics relations to the leptons become more important. For the assignment accuracy, the four leading kinematics variable all relate to the angular features of jets and leptons and show quite some difference to the next important variable, which is the jets' energy. For the selection accuracy, it can be seen that energy and transverse momenta rank higher than the azimuthal angles. This can be explained by considering, that the jets from the signal process tend to be more energetic. For both metrics, global events features, such as variable related to missing transverse energy, the number of jets and angular separation of the leptons appear among the least impactful variables, with the azimuthal angle of the MET leading for the assignment accuracy and the number of jets for the selection accuracy. Lastly, it can be observed that for both metrics, the charge of the leptons seems to be entirely irrelevant. This underlines the previous hypothesis, that the assignment of jets to leptons is largely independent of the lepton charge. This makes permutation invariant models such as the CompactTransformer favourable. Generally, the insights into the model's reliance on input features, confirmed previous expectation. The model seems to utilise the variables in a way, that is consistent with the baseline-methods. Notably, however, the model generally depends only weakly on the MET variables, showing, that information related to the neutrinos seems to have little impact.

6.2. Combination with Neutrino Regression

To achieve a full reconstruction of the dileptonic $t\bar{t}$ events, the jet assignment model can be combined with a neutrino regression model to predict the momenta of the two neutrinos in the event. This combination allows for a more comprehensive understanding of the event kinematics and can improve the overall reconstruction performance. This allows not only to reconstruct the full kinematics of the event, but also to potentially leverage the interdependencies between the jet assignment and neutrino regression tasks to improve the performance of both. For example, the jet assignment model can provide additional context for the neutrino regression model, which can help it make more accurate predictions by considering the relationships between the jets and leptons in the event. Conversely, the neutrino regression model can provide additional information about the event kinematics that can help the jet assignment model make more informed decisions about which jets are associated with which leptons. By combining these two models in a unified framework, it is possible to achieve a more holistic approach to event reconstruction in the dileptonic $t\bar{t}$ channel, which can lead to improved performance and a better understanding of the underlying physics processes.

6.2.1. Unified Model Architecture

To combine the jet assignment and neutrino regression tasks in a unified model architecture, the CompactTransformer (CPT) architecture can be extended by adding a head for neutrino regression. This new architecture, which we refer to as the MultiModalTransformer (MMT), retains the core structure of the CPT while incorporating additional components to handle the neutrino regression task. The only difference stems from the output layers, which now also includes a regression head. This regression head takes the latent vectors of the met features after the shared transformer layers and concatenates them with the latent vectors of the leptons to predict the momenta of the two neutrinos associated with the leptons. The regression head consists of a shared DNN that processes the concatenated features and outputs the predicted neutrino momenta. By sharing the transformer layers between the jet assignment and neutrino regression tasks, the MMT architecture can leverage the interdependencies between these tasks to improve the per-

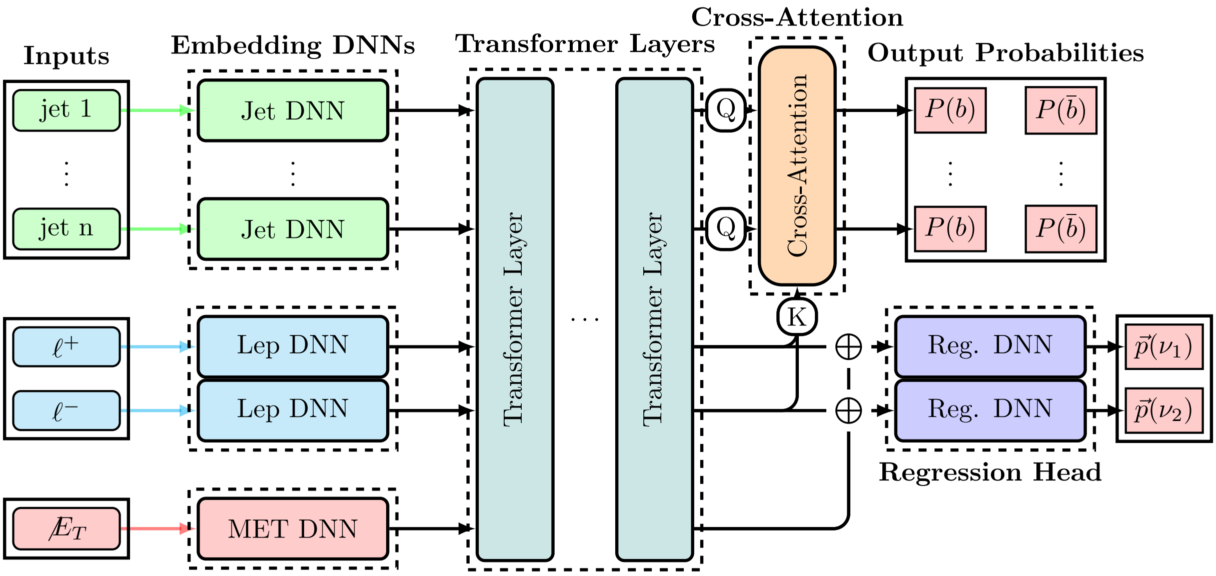


Figure 6.7.: This schematic visualises the general structure of the CompactTransformer architecture equipped with an additional head for neutrino regression. Note, that $\oplus$ represents a concatenation of the corresponding features. This architecture is called MultiModalTransformer (MMT).

formance of both. This unified architecture provides a more holistic approach to event reconstruction in the dileptonic $t\bar{t}$ channel, enabling it to capture the complex relationships between jets, leptons, and neutrinos in a single model. It was also investigated to use the assignment scores from the jet assignment to provide a weighted projection of the jet features to the regression head. The idea was to provide additional context for the regression head by considering the relationships between the jets and leptons in the event. For this, the jet features were weighted by the corresponding assignment scores for each lepton and then concatenated to the features of the leptons and met before being passed to the regression head. This approach, however, did not lead to an increase in regression performance. On the other hand, however, it decreased the accuracy of the jet assignment task. This suggests, that the additional context provided by the weighted jet features may have introduced noise or irrelevant information that negatively impacted the performance of the jet assignment task, while not providing sufficient benefit to the neutrino regression task to justify its inclusion. Thus, it was decided to not include this additional context in the final MMT architecture, as it did not lead to an overall improvement in performance and had a negative impact on the jet assignment accuracy.

6.2.2. Training Procedure

The training setup follows closely the one outlined in Section 6.1.2. The main difference is that the loss function now includes an additional term for the neutrino regression task. The total loss is computed as a weighted sum of the cross-entropy loss for the jet assignment and the loss function for the neutrino regression. For the neutrino regression, different loss functions are investigated. All of them are based on some distance measure between the predicted and true neutrino momenta. An overview of the different loss-functions can be found in Table 6.1. The idea behind the different loss-functions is to reflect that the neutrino regression should be not only guided by the deviation with respect to the components but rather the physical objects these components represent. Different loss-function can thus emphasise different aspects of the neutrino properties, that the transformer should aim to capture correctly. Note, that for all loss-functions the output of the model remain the Cartesian coordinates of the two neutrinos. Only the method to compute the difference between truth and prediction is changed. Designing the most useful loss-function may very well depend on the target of one's analysis. Fulling studying how to tune the difference measure exceeds this study. Thus, only three different instances are evaluated here to showcases the impact of these. The Cartesian-Loss computes just a regular MSE of the Cartesian coordinates of the neutrino momenta. In contrast, the PtEtaPhi-loss measures the difference between the true and predicted neutrino momenta in terms of their p_T, η, ϕ coordinates. Lastly, the MagnitudeDirection-loss quantises the deviation from the truth as the MSE of the vector's magnitude and cosine similarity. For this Chapter, however, the loss is taken as the mean-squared error (MSE) for each neutrino component. That is, because for investigating the architecture and its scaling the explicit loss-function was not found to have a significant impact. Apart from the loss function, the weighting of the two tasks and the scaling of the regression targets also have to be tuned. Here, it was found to be beneficial to scale the regression targets to units of 100 GeV. This helps to improve the training stability and convergence. The weighting was, after some experimentation, set to $w_{assignment} = 1$ and $w_{regression} = 0.2$, which provided a good balance and stable training behaviour. All other parameters of the training procedure, such as the optimizer, learning rate scheduling, and early stopping criteria, are kept the same as in the jet assignment training procedure to ensure consistency and allow for a fair comparison between the models. During the training process, the performance of both the jet assignment and neutrino regression tasks is monitored on the validation set to ensure that the model is learning effectively and not overfitting to the training data. To evaluate the performance of the combined model, separate metrics for both tasks are reported during the training process. For the jet assignment, the accuracy of the assignments is monitored, while for the neutrino regression, metrics such as relative error of the predicted neutrino momenta compared to the true values are tracked. The relative error is computed as the L^2-Norm of the difference between the predicted and true neutrino momenta, divided by the L^2-Norm of the true neutrino momenta, which provides a measure of the accuracy of the predictions relative to the scale of the true values.

Loss	Computation
Cartesian-Loss	$L = (\Delta p_x)^2 + (\Delta p_y)^2 + (\Delta p_z)^2$
PtEtaPhi-Loss	$L = (\Delta p_T)^2 + (\Delta \eta)^2 + (\Delta \phi)^2$
MagnitudeDirection-Loss	$L = (\Delta \lvert \vec{p} \rvert)^2 - \frac{\vec{p}_{pred} \cdot \vec{p}_{true}}{\lvert \vec{p}_{pred} \rvert \lvert \vec{p}_{true} \rvert}$

Table 6.1.: Overview of different loss-function that were investigated.

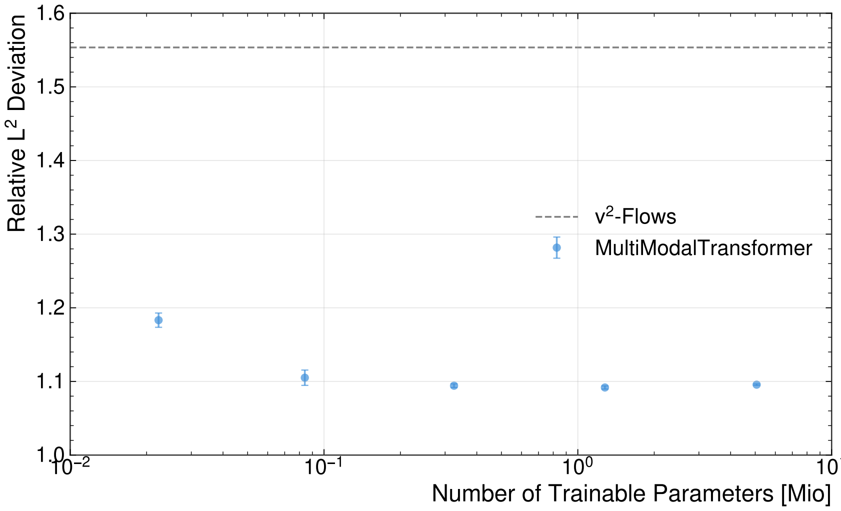
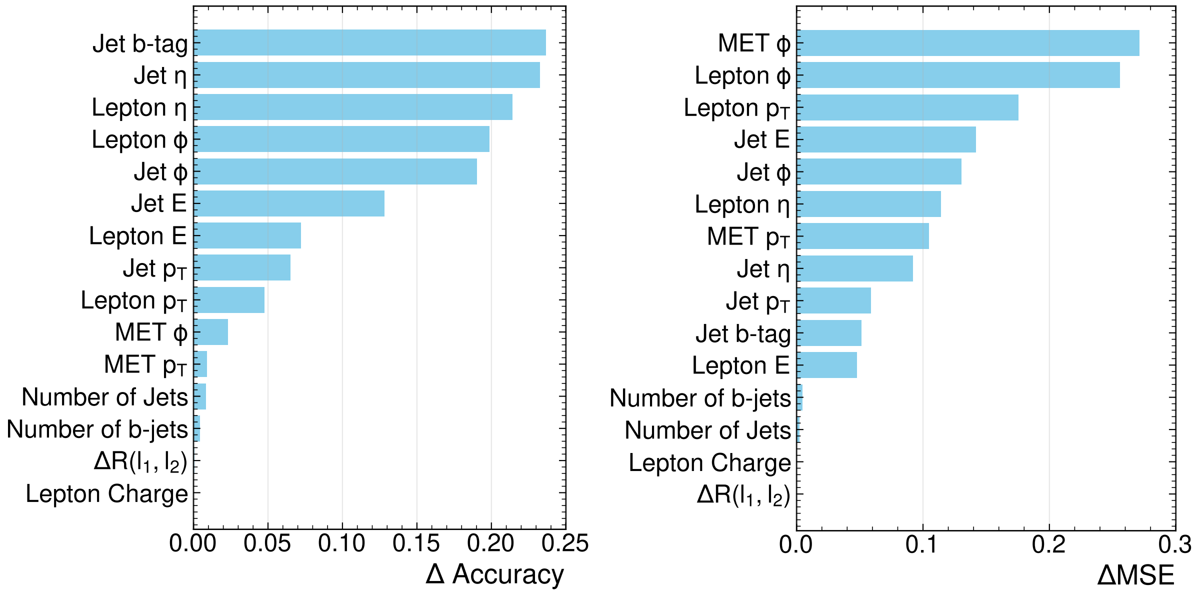


Figure 6.8.: Relative L^2 deviation of the predicted neutrino momenta compared to the true values as a function of the number of trainable parameters for the MultiModalTransformer (MMT) architecture.

6.2.3. Performance Evaluation

As the evaluation for the assignment models showed very minor difference in performance from different numbers of layers, for the combined model, the number of layers is kept fixed to 6, while the embedding dimension is varied in the range of $\{16, 32, 64, 128, 256\}$ to evaluate its impact on the performance of both tasks. Due to the increased complexity of the combined model, it is expected that a higher embedding dimension may be required to achieve good performance on both tasks simultaneously. To keep the computational resources feasible, it was decided to not perform a full grid search for the combined model, but rather to focus on evaluating the impact of the embedding dimension while keeping the number of layers fixed. The Figure 6.8 shows the performance of the MMT architecture in terms of the relative L^2-Norm deviation of the predicted neutrino momenta compared to the true values as a function of the number of trainable parameters. For comparison, the performance of the neutrino regression provided by ν^2-Flows is plotted as a horizontal line, which serves as a baseline for evaluating the performance of the MMT architecture. From the results, two main observations can be made. First, the performance of the MMT architecture generally seems to outperform the ν^2-Flows baseline across different embedding dimensions, which suggests that the model architecture can effectively leverage the interdependencies between the jet assignment and neutrino regression tasks to improve the performance of both. Second, there is a trend of improving performance with increasing number of trainable parameters, which indicates that the model benefits from having sufficient capacity to learn the underlying patterns in the data. However, it is also observed that the performance improvement tends to plateau for higher embedding dimensions, which may suggest that there is a point of diminishing returns where increasing the model size further does not lead to significant performance gains. Since no decrease in performance for higher number of parameters was found, despite being trained with the number of events, the model does not seem to show effects of overfitting for these model sizes. This is most likely to be attributed to the aggressive dropout and regularisation applied during the training.

6.2.4. Feature Importance



(a) Assignment Accuracy **(b)** Regression MSE

Figure 6.9.: The Plots (a) and (b) show the difference in assignment accuracy and regression MSE, respectively, of the CPT-model when permuting the input features. For each feature, five different permutation are performed. The features are ranked in decreasing order of the observed reduction. Note, that the mean-square error is again computed as the relative deviation of the L^2-norm as before.

Analogous to the models used for assignment, the model's reliance on the input features is investigated using permutation importance. The results are shown in Figure 6.9. Despite, the model being trained with the additional training target, the feature importance scores for the assignment accuracy have not changed at all and match close to perfect with those in Figure 6.6. The feature importance for the regression performance, however, shows a very different dependence on the input features. On one hand, the jets' b-tag, which was is the leading feature for the assignment, only has a very little impact and scores lower than all kinematic features except the energy of the leptons. Generally, it can be observed, that while the pseudorapidity scores higher for the assignment than the azimuthal angles, this is completely reversed for the regression, where both the leptons' and jets' ϕ scores higher than their η. Also, features related to the leptons' energy, such as their transverse momenta, become much more important. In stark contrast to the assignment's feature importance, for the regression, the azimuthal angle ϕ of the MET manifests as the most importance variable, closely followed by the lepton's azimuthal angle. Given that the direction of the MET is very strongly related to the neutrinos' momenta, this is unsurprising. Interestingly, the absolute value of the MET scores rather low and seems to have much less impact. It should be noted at this point that the scoring of the feature is expected to strongly depend on the metric, employed to measure model's performance in the given task. While this was not further investigated for this thesis, all results for the feature importance of the regression are to be taken with a grain of salt. But the general observation that the features differ a lot compared to the assignment task, remains noteworthy. Since the b-tag is kinematically not related to the neutrinos, its non-zero score implies, that the regression does depend on some internal assignment of the jets within the transformer's latent-space.

6.2.5. Multitask Learning Benefits

One of the key prospects of combining the jet assignment and neutrino regression tasks in a unified model architecture is the potential for multitask learning benefits. The idea is, that the information exchange and additional context provided by the shared transformer layers can help both tasks to learn more effectively from the data, and thus increasing the overall performance of the model. To investigate this, the same model architecture is trained separately for three different setups:

1. CompactTransformer: a MultiModelTransformer trained only for the jet assignment task, with a loss function that includes only the cross-entropy loss for the jet assignment and no contribution from the neutrino regression task ($w_{assignment} = 1$ and $w_{regression} = 0$).

2. MultiModalTransformer: a MultiModelTransformer trained for both the jet assignment and neutrino regression tasks, with a loss function that includes both the cross-entropy loss for the jet assignment and the mean-squared error loss for the neutrino regression, weighted by $w_{assignment} = 1$ and $w_{regression} = 0.2$.

3. RegressionTransformer: a MultiModelTransformer trained only for the neutrino regression task, with a loss function that includes only the mean-squared error loss for the neutrino regression and no contribution from the jet assignment task ($w_{assignment} = 0$ and $w_{regression} = 1$).

The results of this comparison are shown in Figures 6.10 and 6.11. They compare the regression accuracy and assignment between MultiModalTransformer with the Compact-Transformer and RegressionTransformer, respectively, to evaluate the benefits of multitask learning in the unified model architecture. From this, it can be seen that the MultiModalTransformer shows no improvement in the jet assignment accuracy compared to the CompactTransformer, which suggests that the inclusion of the neutrino regression task does not provide significant benefits for the jet assignment task. However, it can be observed that the MultiModalTransformer shows an improvement in the neutrino regression performance compared to the RegressionTransformer, which indicates that the inclusion of the jet assignment task provides valuable context and information that helps the model to make more accurate predictions for the neutrino momenta. Notably, this advantage particularly appears for higher embedding dimensions, which indicates that the embedding dimension has to be large enough to fit the relevant information for both tasks. The impact of the included jet assignment on the regression performance is also indicated by the feature importance as discussed in the previous Section 6.2.4, where it was pointed out that the non-vanishing impact of the b-tagging on the regression targets, show, that the model does require an internal assignment of the jets to perform the regression. Overall, these results highlight the potential advantages of multitask learning in this context and suggest that combining related tasks in a unified model architecture can lead to improved performance for certain tasks by leveraging shared information and context between them. In particular, the jet assignment, which was often overlooked in previous works such as ν^2-Flows, can provide valuable context for the neutrino regression task, which can lead to significant improvements in its performance when combined in a unified model architecture. Furthermore, it should be mentioned, that the computational cost of training the MultiModalTransformer is not significantly higher than training the CompactTransformer or RegressionTransformer separately, as the shared transformer layers allow for efficient learning of both tasks without requiring a substantial increase in model complexity. Similar arguments hold for the inference time, which is also not significantly higher for the MultiModalTransformer compared to the separate models, as the shared architecture allows for efficient processing of both tasks without a substantial increase in computational requirements. As the computational cost for the ν^2-Flows method could not be evaluated, a direct comparison of the different methods in terms of computational efficiency is not possible. But in general, the benefits of multitask learning in terms of improved performance can be achieved without incurring significant additional computational costs, making it an attractive approach for event reconstruction in the dileptonic $t\bar{t}$ channel. On the other hand, training and running separate models for both tasks can be expected to double the computational cost, as both models would require separate forward passes and training processes. To conclude, while the combination of both tasks in one model did not provide enhancements in the assignment performance as expected, it can be said that the performance of the region does significantly increase if assignment is added as a training target. Furthermore, the model can perform both tasks without the need for a larger embedding dimension. That indicates, that there is some information sharing and synergy in the embedding and the model does not require additional dimensions to capture both. Thus, a model can be extended to perform the full reconstruction instead of only one singular tasks while maintaining its size and run time. This can significantly decrease computational costs for training and inference. For the analyses of data taken during the High Lumi era of the LHC, this is a very important consideration as the computational load will increase considerably due to larger datasets. In general, balancing the computational cost will become much more relevant for upcoming analyses, and approaches that can reduce computational overhead will be favourable.

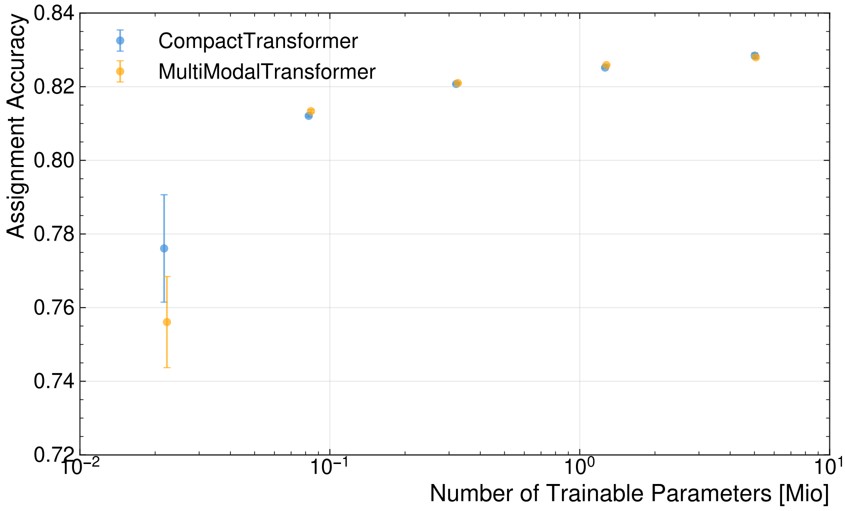


Figure 6.10.: Assignment Accuracy of the MultiModalTransformer trained for both the jet assignment and neutrino regression tasks compared to CompactTransformer, which features the same architecture trained only for the jet assignment task, as a function of the number of trainable parameters.

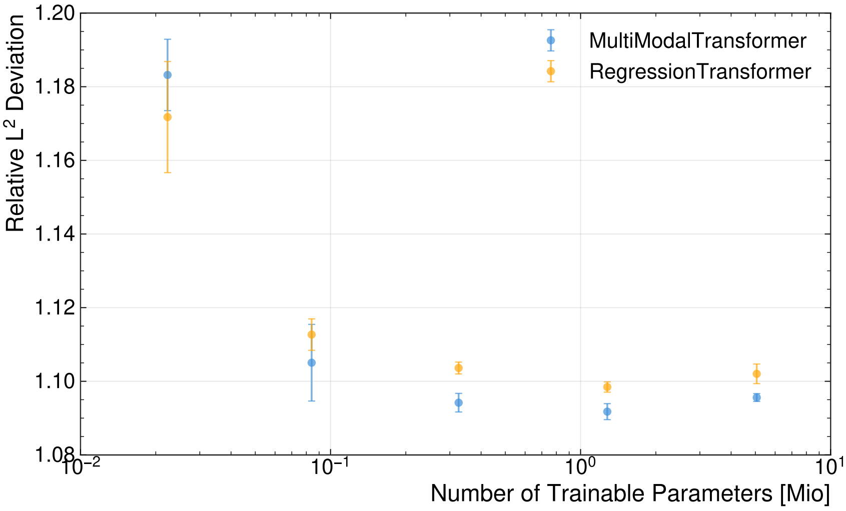


Figure 6.11.: Relative L^2 deviation of the predicted neutrino momenta compared to the true values for the MultiModalTransformer and the RegressionTransformer as a function of the number of trainable parameters.

7. Evaluation of the ML-Based Reconstruction

To evaluate the performance of the ML-based reconstruction method, several metrics and analyses are conducted. This chapter presents the results of these evaluations, focusing on the accuracy of particle assignments, the reconstruction of relevant physics variables, and the performance across different kinematic regions. For this evaluation, the ML-based methods are compared against the baseline reconstruction techniques. The comparison includes both the overall performance and the performance in specific regions of interest, such as near the $t\bar{t}$ production threshold. Furthermore, the impact of the improved reconstruction on the resolution of key observables is assessed, demonstrating the potential benefits for precision measurements and searches in top-quark physics.

7.1. Assignment Accuracy

Method	Assignment Accuracy	Selection Accuracy
$\Delta R(\ell, j)$-Method	$0.63455^{+0.00050}_{-0.00042}$	$0.92333^{+0.00035}_{-0.00041}$
χ^2-Method	$0.73896^{+0.00039}_{-0.00048}$	$0.92329^{+0.00031}_{-0.00037}$
CompactTransformer	$0.8230^{+0.0005}_{-0.0006}$	$0.95946^{+0.00024}_{-0.00036}$

Table 7.1.: Comparison of assignment and selection accuracies for different reconstruction methods. The uncertainties represent the statistical variations in the evaluation dataset.

The assignment accuracy is defined as the fraction of events where the reconstructed particles are correctly assigned to their parent partons. The selection accuracy, on the other hand, measures the fraction of events where the correct assignment is among the selected candidates. Both metrics are crucial for evaluating the performance of reconstruction methods, as they directly impact the resolution of physics observables and the sensitivity of analyses. Furthermore, some variables, such as the invariant mass of the $t\bar{t}$ system, are only sensitive to the correct selection of candidates, making the selection accuracy particularly important for certain analyses. The Table 7.1 summarises the overall assignment performance of the reconstruction methods. The ML-based method, represented by the CompactTransformer, shows a significant improvement in the assignment accuracy compared to the $\Delta R(\ell, j)$-Method and the χ^2-Method. Concerning the baseline methods, the χ^2-Method performs roughly 16% more accuracy than the $\Delta R(\ell, j)$-Method. The difference between the two baseline methods is even slightly larger than the difference between the CompactTransformer and the χ^2-Method. For the selection accuracy, both baseline methods yield the same result within the uncertainties, as they make use of the same selection criteria of the jets. The improvement of the ML-based method for the selection accuracy is rather small, which can be attributed to the b-tagging. Because baseline methods almost exclusively rely on b-tagging for the selection, and the GN2-tagger is very performant [65], the selection accuracy is already very high. Further, the b-tagging of the GN2-tagger already takes into account the full kinematic information of the event, making it difficult for the ML-based method to significantly outperform the selection accuracy. This is also supported by the model's reliance on the input features in Section 6.1.4. The investigation revealed that the selection accuracy mainly depends on the b-tag. It does however utilise some information provided by the kinematics of the jets and leptons, giving it an edge with respect to the baseline methods, which only rely on b-tagging and jet p_T. Regarding the assignment accuracy, the ML-based method shows a significant improvement, which can be attributed to its ability to learn complex correlations between the kinematic variables of the reconstructed particles. This complex correlations can be much more efficient than the simply criteria of the baseline methods, allowing to surpass their performance.

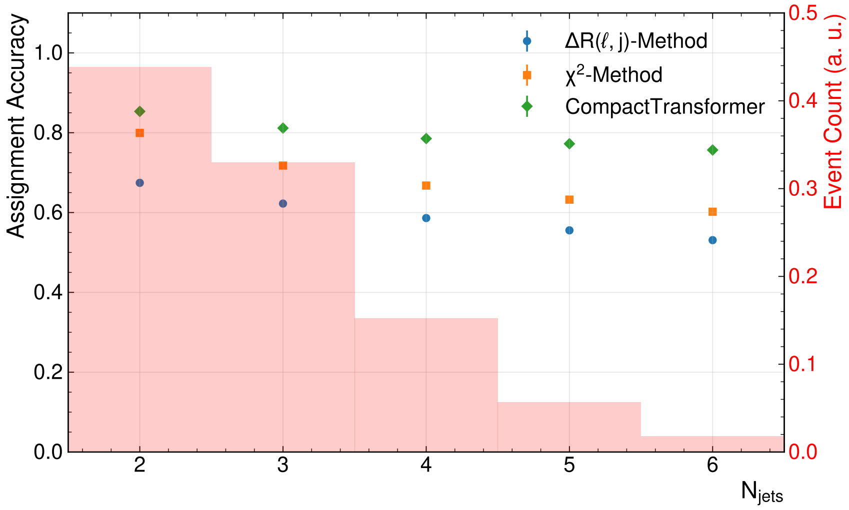


Figure 7.1.: Assignment accuracy as a function of the number of jets in the event for different reconstruction methods. The background histograms show of the distribution of events as a function of the number of jets. The uncertainties represent the statistical variations in the evaluation dataset.

The Figure 7.1 shows the assignment accuracy for events with different jet multiplicities for each method. It can be observed, that the CompactTransformer outperforms the baseline-methods in all bins. For all methods, the performance drops with an increasing number of jets. The performance drop, however, is relatively small, which can be attributed to the performance of the b-tagging. This is in stark contrast to the methods for the all-hadronic channel, where the performance drops very strongly for events with many jets in the final-state as seen in Reference [6]. Even with additional jets, the chance of a jet being falsely b-tagged is quite low [65]. Thus, the performance of selecting the correct jets degrades much less compared to the hadronic channels, where also non-tagged jets have to be selected. Notably, the performance of the CompactTransformer drops the least, and the difference in performance increases for higher jet numbers. This stems from the improved accuracy of the jet-selection as shown before. In the Appendix, the Figure A.14 shows the selection accuracy binned in the same way. It can be seen that the CompactTransformer does a much better job at selection the correct jets for events with a higher jet multiplicity. It was found, that this is the primary cause for differences in assignment accuracy with respect to the jet-multiplicity. The efficiency of assigning correctly selected jets seems to be mostly independent of the jet-multiplicity. Figure 7.2

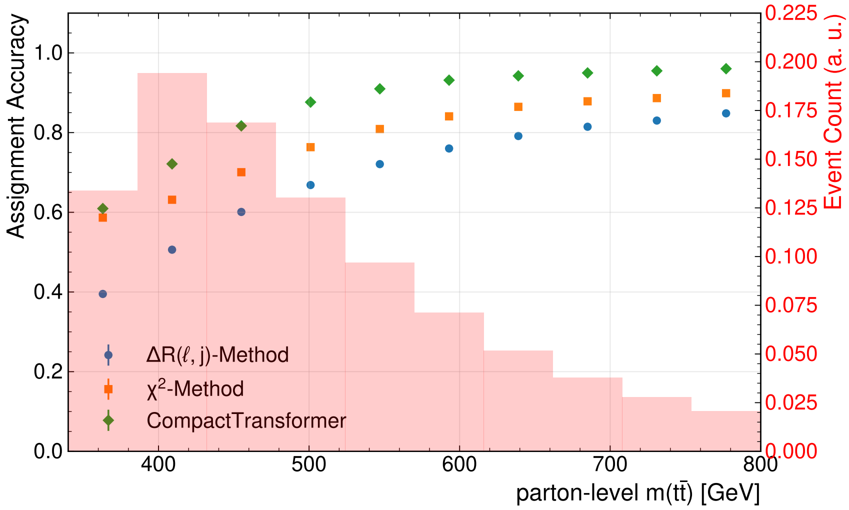


Figure 7.2.: This plot shows the assignment accuracy in bins of the true invariant mass of the top-quark pair.

shows the assignment accuracy as a function of the parton-level $m(t\bar{t})$. All methods show a similar behaviour; the accuracy drops for low invariant masses and seems to saturate for high invariant masses. For the CompactTransformer, the saturated performance in the high invariant mass regime almost approaches unity, marking near perfect reconstruction. On the other end of the spectrum, the performance drops to around 0.6 near the $t\bar{t}$-production threshold. For these low-mass events, the CompactTransformer and the χ^2-Method appear to be almost on-par. In general, this indicates that the χ^2-Method is the least sensitive to $m(t\bar{t})$, while the $\Delta R(\ell, j)$-Method seems to be most sensitive. As it was found before, the number of jets can impact the accuracy. Interestingly, the double-binned distribution of events with respect to the invariant mass $m(t\bar{t})$ and number of jets in Figure A.3 shows that events with lower energy have fewer jets on average. Naively, one would expect this to increase the assignment accuracy, which is the opposite of what is observed. To decouple the number of jets per event and the selection accuracy altogether, one can consider the conditional probability to make a correct assignment, given that the correct jets were selected. For this, the assignment reduces to a binary problem in the dilepton case, as jets can only be assigned to bottom- or antibottom-quark respectively. Thus, the probability for randomly assigning correctly, is at 50%. According to the Bayes' theorem, the conditional probability for the different methods can be computed by dividing the assignment accuracy with the selection accuracy. The corresponding plot showing

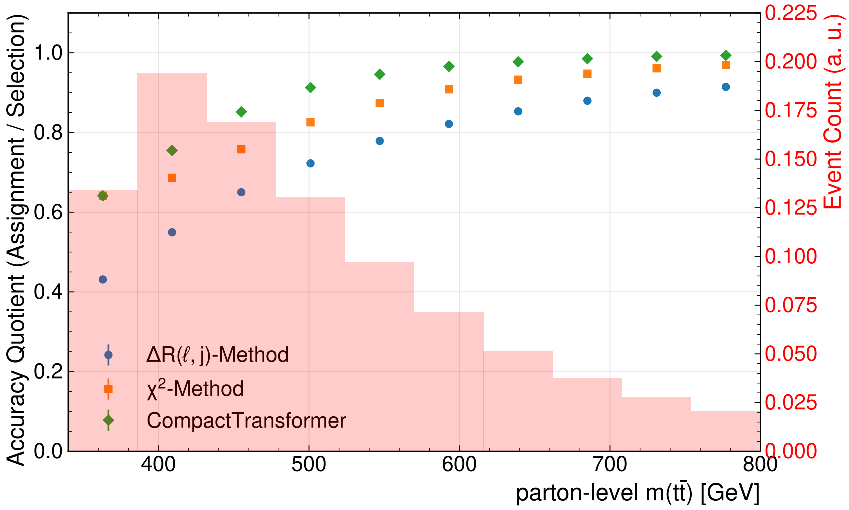


Figure 7.3.: Quotient of assignment and selection accuracy computed in bins of the parton-level $m(t\bar{t})$.

this probability for the different methods, binned in parton-level $m(t\bar{t})$, is displayed in Figure 7.3. This investigation allows to separate the process of actually assigning the selected jets. One thing one to note, is that the $\Delta R(\ell, j)$-Method has a performance below 0.5 for the lowest bin, meaning that it performs worse than a random assignment in this bin. This suggests, that the topology of the particles in this region actually disfavours assigning jets based on proximity. Figure A.6 in the Appendix clearly shows the difference in angular distance distributions for correct and incorrect jet-lepton pairing between the inclusive sample and the near-threshold region. This serves as an explanation for the significant performance drop-off in this region. Going further, this behaviour can be directly linked to the underlying physics: At the production-threshold, the top quarks carry very little energy. Thus, the decay products of the top quarks, i.e. the jets and W-boson and subsequently the lepton are less collimated. This is because the bottom-quark and W-boson are back-to-back in the top quark's rest frame and get collimated by the Lorentz boost of the top quark with respect to the lab frame. The strength of this boost is directly proportional to the top quark's energy, which itself is (approximatively) proportional to the $m(t\bar{t})$. Note, that this picture get more complicated if one takes the chiral-nature of the decays into account, which can effect the angular distributions. As the energy increases, decay objects become increasingly collimated. While this serves to explain why the performance drops, it does not explain how the performance can be worse than randomly assigning jets. To explain this effect more thoroughly investigations into the angular structure of the events in this region would necessary. Another observation is, that in the lowest bin, the CompactTransformer and χ^2-Method are completely on par for this metric. Performance losses of the chi^2-Method can be explained by the Figure A.7 in the Appendix, which shows the different distributions of the reconstructed top-mass for the correct and incorrect assignment of the jets. For comparison, the distributions for the near-threshold and for the inclusive sample are shown. The correct pairing shows the typical Breit-Wigner mass-peak, one expects, while the incorrect pairs have a washed out distribution. For the low-energy events, the difference between the two distributions is smaller but still very pronounced. One reason is that the top quarks tend to be more off-shell, and faulty assignments do not change the kinematics as dramatically. But, generally, the assignment based on the invariant mass seems to be relatively robust. For the machine-learning methods, it is difficult to assess which particular features lead to the degrading of the performance. The leading feature, i.e. the b-tag, is not found to have a varying performance in this range of invariant mass $m(t\bar{t})$ - as seen in Figure A.2. For the other features it cannot be easily assessed how their distributions in the different regimes effect the assignment performance. One way, is to see if certain event features of near-threshold events impact the performance. A particular example for this is the angular separation $\Delta(\ell_1, \ell_2)$ of the leptons. The property of having less energy - and thus less boosted - top-quarks, also significantly impacts the angular separation of the leptons themselves. While decays of high-energy top-quark pairs lead to back-to-back leptons, the opposite is true for low-energy top-quark pairs. The double-binned distribution of events in Figure A.5 shows that events with lower $m(t\bar{t})$ tend to have more collimated leptons, whereas they are more separated for more energetic events. This confirms the expectation, that events at the threshold have closer leptons. Since the angular separation of the leptons is suspected to worsen the performance of the events, its impact on the individual methods is investigated. As before, Figure 7.4 shows the assignment accuracy

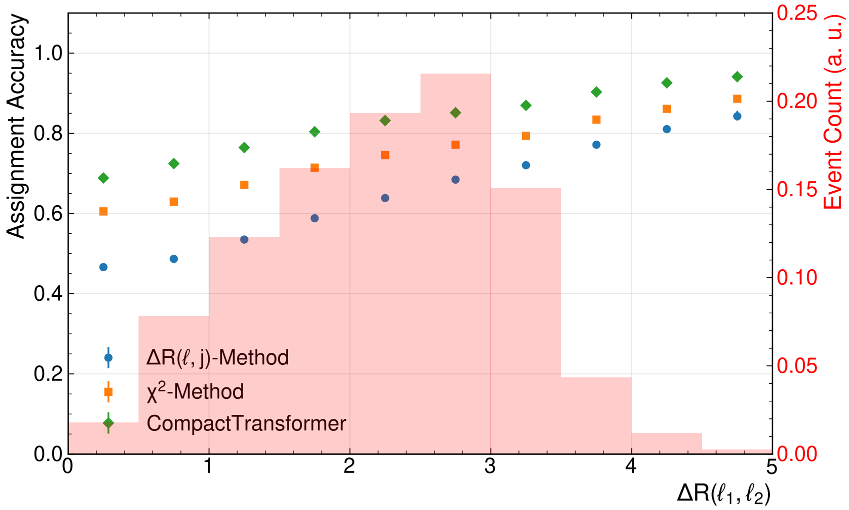


Figure 7.4.: Assignment accuracy binned in the angular distance between the two leptons of the events $\Delta R(\ell_1, \ell_2)$.

in bins of the angular distance $\Delta R(\ell_1, \ell_2)$ between the two leptons. All methods suffer losses in performance for events with close leptons. Interestingly, however, for the smallest[1] angular separation, the performance is still above the one for near-threshold events. This implies, that the decreased performance due to the reduced angular separation in the near-threshold region is one contributing factor, but not the only one. A combined comparison of both variables' impact is shown in Figure A.12 in the Appendix. This

[1]Due to the overlap-removal, which removes overlapping objects from the (simulated) events, there is a cut-off at $\Delta R < 0.3$. This technique accounts for the fact that in a detector objects that coincidence cannot be separated.

distribution provides a more comprehensive picture, as it is able to separate the effect of both variables. Also, it allows to isolate the effects of both, by considering a variation of one variable while keeping the other fixed. The double-binned accuracy reveals, that the invariant mass $m(t\bar{t})$ seems to have a much stronger impact on the performance, than the angular separation alone. Expectedly, the $\Delta(R, j)$-Method is the most sensitive to the angular separation. For the lowest bins in $\Delta R(\ell_1, \ell_2)$, the performance breaks down significantly and is nearly constant for varied $m(t\bar{t})$. The other methods' performances reduce much more in the rows of varied $m(t\bar{t})$, than they do in the columns of varied $\Delta R(\ell_1, \ell_2)$. This observation holds for all bins of energies and angular separations considered here. From this, it can be concluded, that the near-threshold region introduces special challenges for assigning the jets. For the traditional methods, these can be explained quite well, but even the transformer, which is able to leverage very complex correlations between variables, only performs slightly better than the baseline methods. While some impact factors could be identified, i.e. angular separation of the leptons and off-shell tops, the complete picture why this region is particularly challenging remains elusive. Due to the increased efforts to study quantum effects or quasi-bound state effects at the production threshold, this remains a very relevant question to be addressed by future investigations. While it was found, that other kinematic variables have regimes with quite different performance in the assignment accuracy, none had an impact as severe as the invariant mass $m(t\bar{t})$. One to mention here, is the transverse momenta of the top-quark pair system. This variable is very sensitive to initial state radiation. Thus, a high p_T is often accompanied by increased number of jets, as seen in Figure A.4. However, it was found, that the $p_T(t\bar{t})$ seems to have no impact on the accuracy of the ml-based assignment, while it degrades both the baseline-methods. The corresponding binned accuracy evaluation is can be seen in Figure A.11 in the Appendix. This shows, that the transformer seems to be particularly insensitive to initial state radiation compared to the other methods. Another object of study were the aforementioned spin-sensitive variables. As these have been used in recent analyses, it is interesting to study whether particular regions are more or less sensitive to misassignments. To mimic the definition of signal regions used in [2], a double-binned plot, resembling the different signal regions in $c_{\text{hel}}$and c_{han}, shows the assignment accuracy in Figure A.13. While all methods are slightly sensitive to these variables, no major differences are found in these bins. What can be seen, however, is that the methods show slightly varying dependencies on the variables. All methods increase in performance for higher values of c_{hel}, but for $c_{\text{han}}$the $\Delta R(\ell, j)$-Method becomes more accurate at higher values, while for the other methods the opposite is true.

7.2. Neutrino Regression

This chapter will analyse and compare the neutrino regression of the MultiModalTransformer with respect to existing baseline methods. Different metrics will be used to assess the quality of the method.

Method	MAE (p_x) [GeV]	MAE (p_y) [GeV]	MAE (p_z) [GeV]
ν^2-Flows	$33.744^{+0.029}_{-0.018}$	$33.736^{+0.026}_{-0.033}$	$88.74^{+0.06}_{-0.12}$
MultiModalTransformer	$24.285^{+0.016}_{-0.020}$	$24.268^{+0.013}_{-0.016}$	$64.277^{+0.057}_{-0.062}$

Table 7.2.

7.3. Variable Reconstruction

7.4. Region Specific Performance

8. Data-Driven Validation Studies

8.1. Run 2 Data Sample

8.2. Analysis Setup

8.3. Results

9. Conclusion

9.1. Summary

9.2. Outlook

A. Additional Plots

A.1. Data Feature Plots

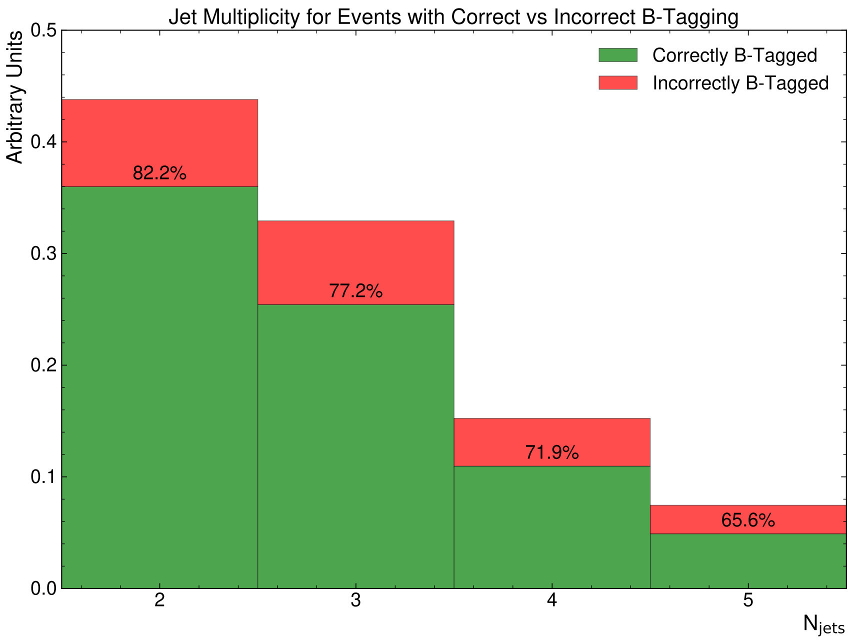


Figure A.1.: This plot shows the b-tagging performance for different numbers of jets in the event. For each jet multiplicity, the number of events, with only jets from top-decays being b-tagged and the number of events with additional or missing b-tagged jets is stacked. The percentage shows the fraction of events with only jets from top-decay being b-tagged.

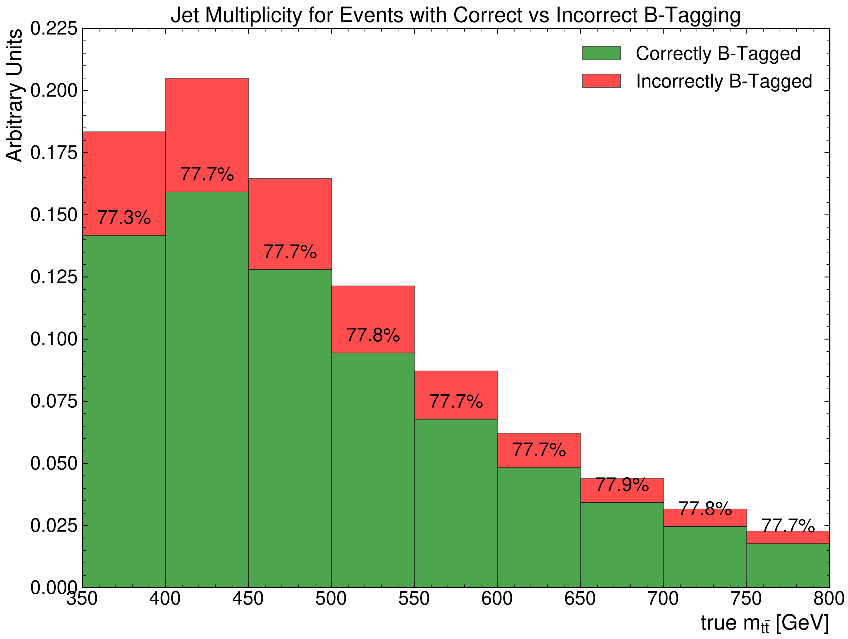


Figure A.2.: This plot shows the b-tagging performance for different bins of the invariant mass $m(t\bar{t})$ at parton-level.

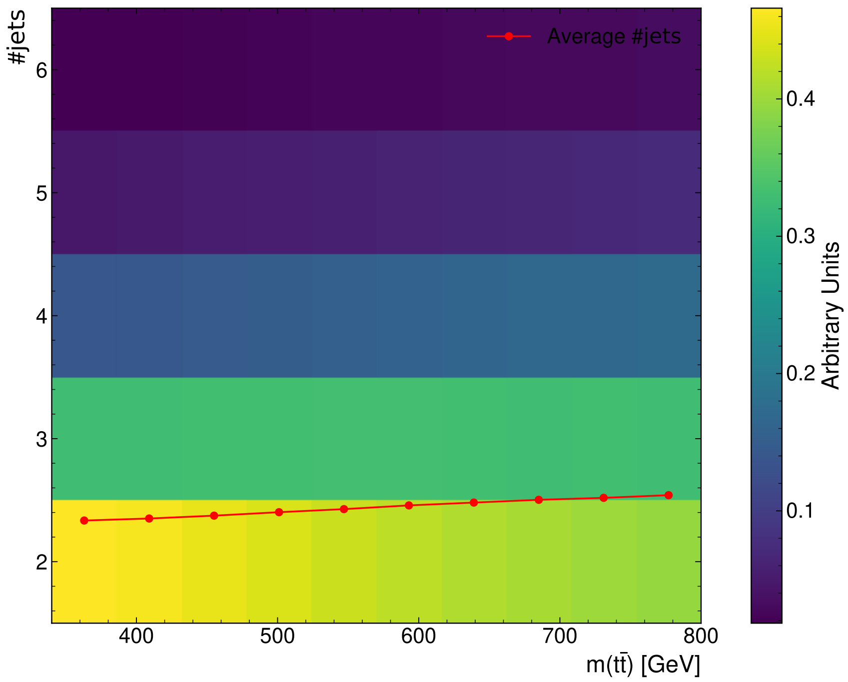


Figure A.3.: This plot shows the b-tagging performance for different bins of the invariant mass $m(t\bar{t})$ at parton-level. For each bin, the number of events, with only jets from top-decays being b-tagged and the number of events with additional or missing b-tagged jets is stacked. The percentage shows the fraction of events with only jets from top-decay being b-tagged.

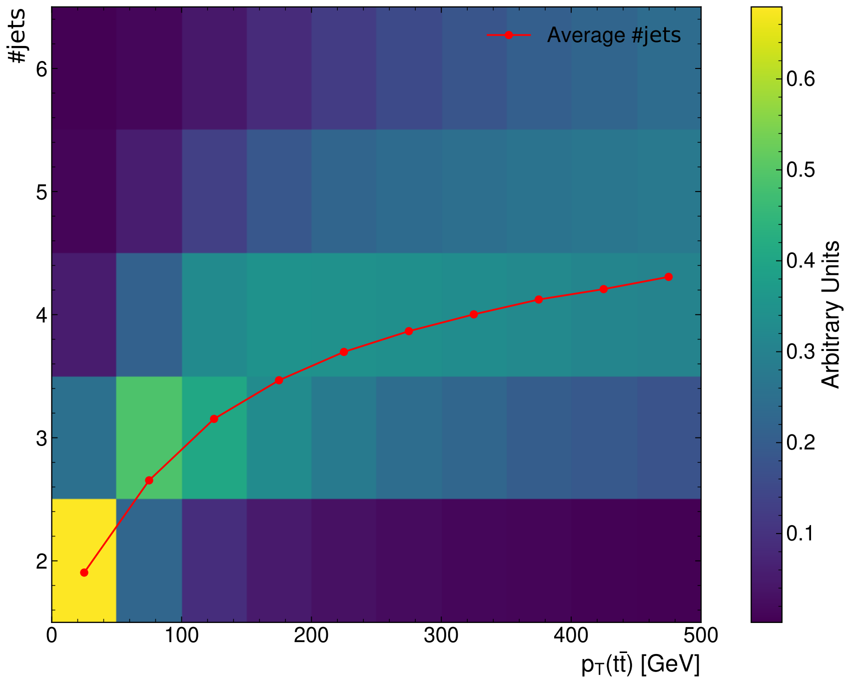


Figure A.4.: This plot shows the distribution of events binned in the parton-level $p_T(t\bar{t})$ and number of jets per event.

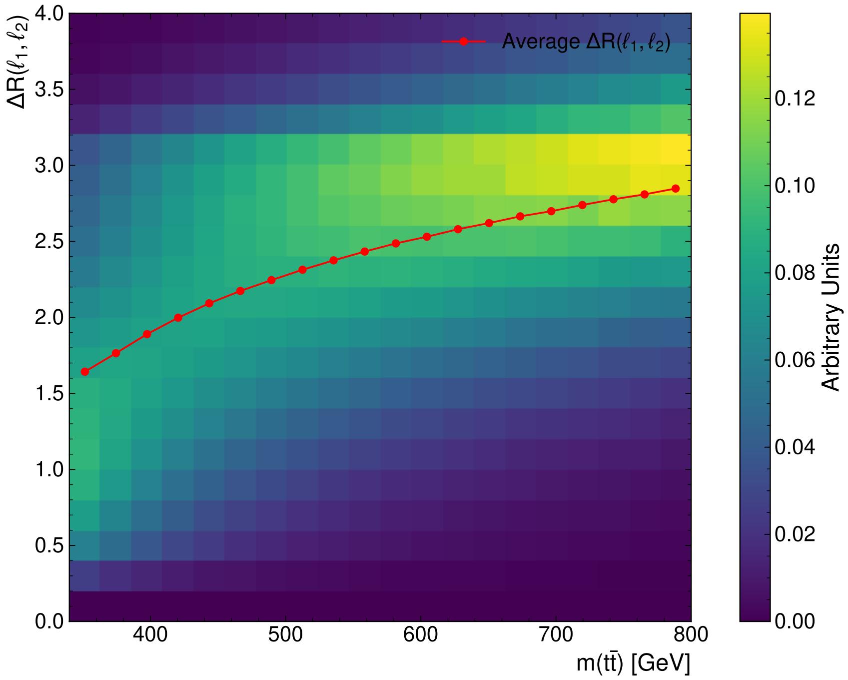
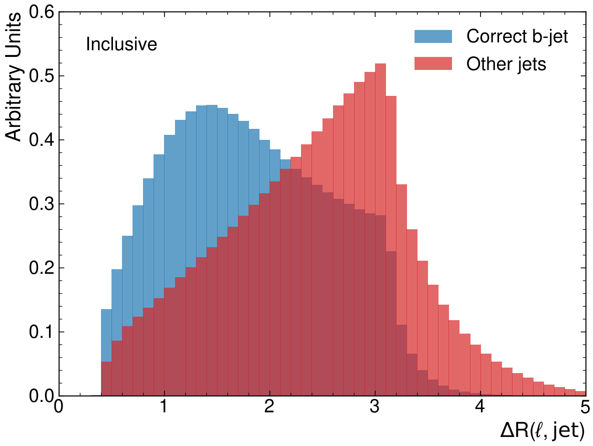
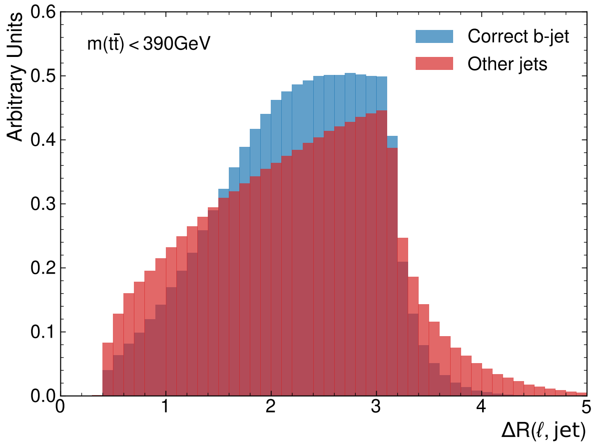


Figure A.5.: This plot shows the distribution of events binned in the parton-level $m(t\bar{t})$ and the angular distance $\Delta R(\ell_1, \ell_2)$.

Threshold Topology

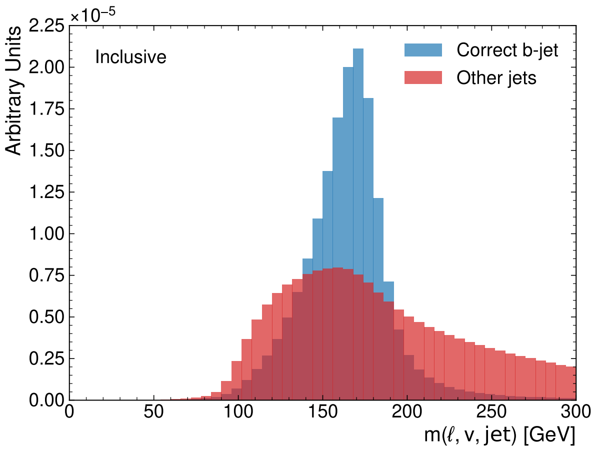


(a) Inclusive

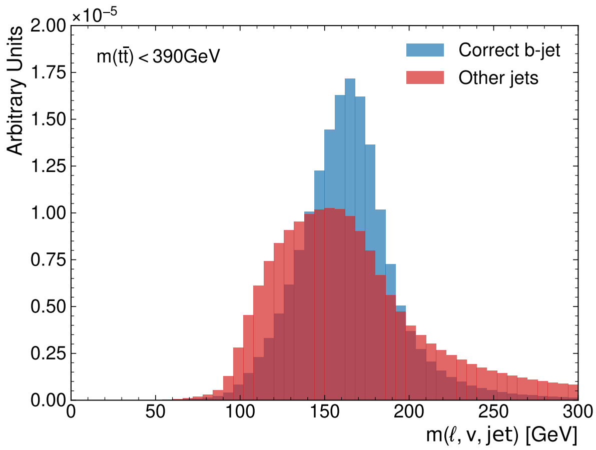


(b) Near-Threshold

Figure A.6.: Distribution of $\Delta R(\ell, j)$ for correct and incorrect pairing of jets and leptons. Evaluated on the full sample in (a) and near-threshold in (b).



(a) Inclusive

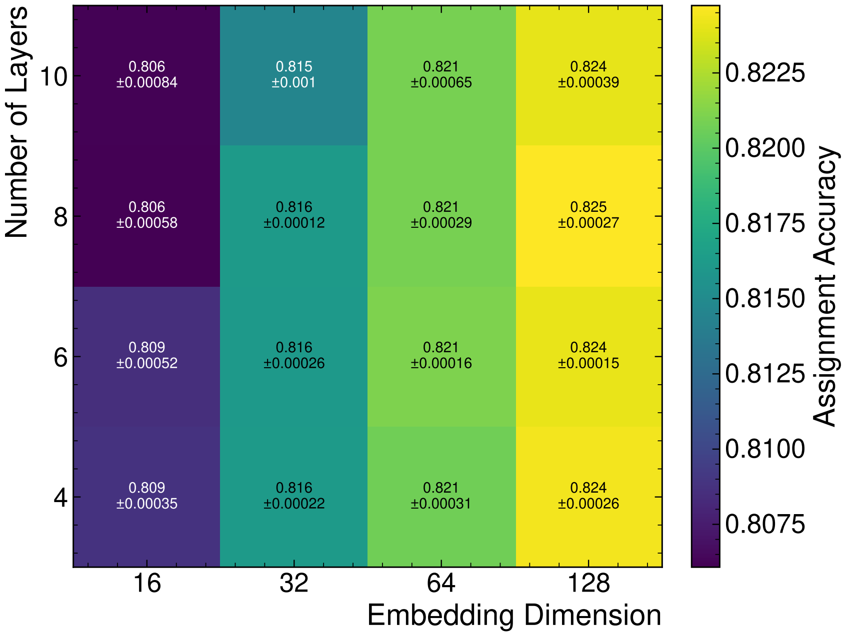


(b) Near-Threshold

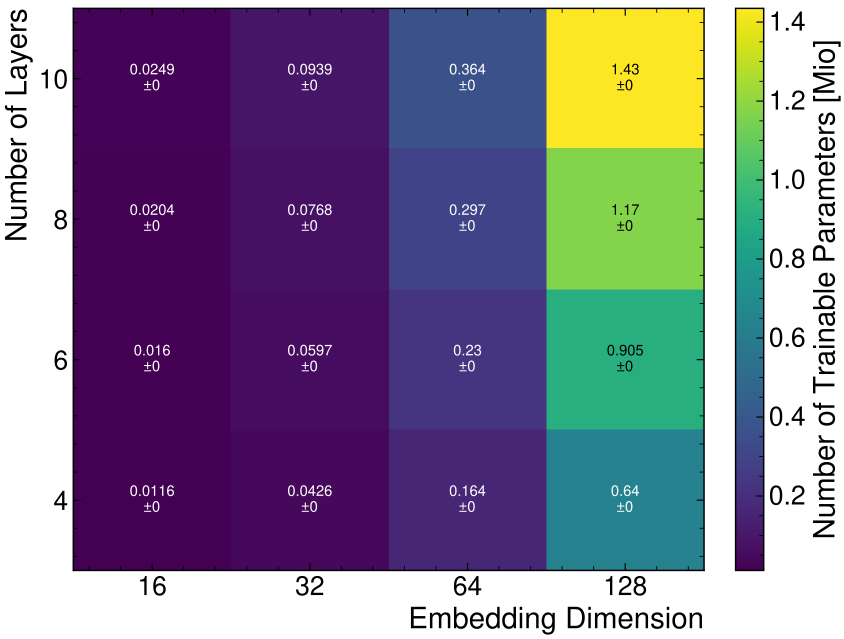
Figure A.7.: Distribution of $m(\ell, \nu, j)$ for correct and incorrect pairing of jets and leptons. Evaluated on the full sample in (a) and near-threshold in (b). Note, that the regressed neutrino momenta from nu^2-Flows were used.

A.2. Hyperparameter Optimisation Results

A.2.1. FeatureConcatTransformer



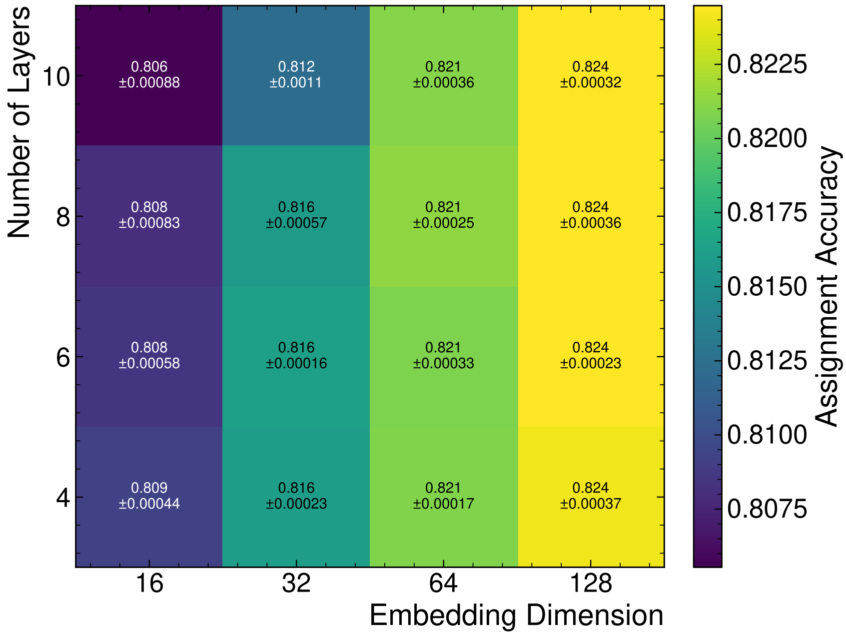
(a) Assignment Accuracy



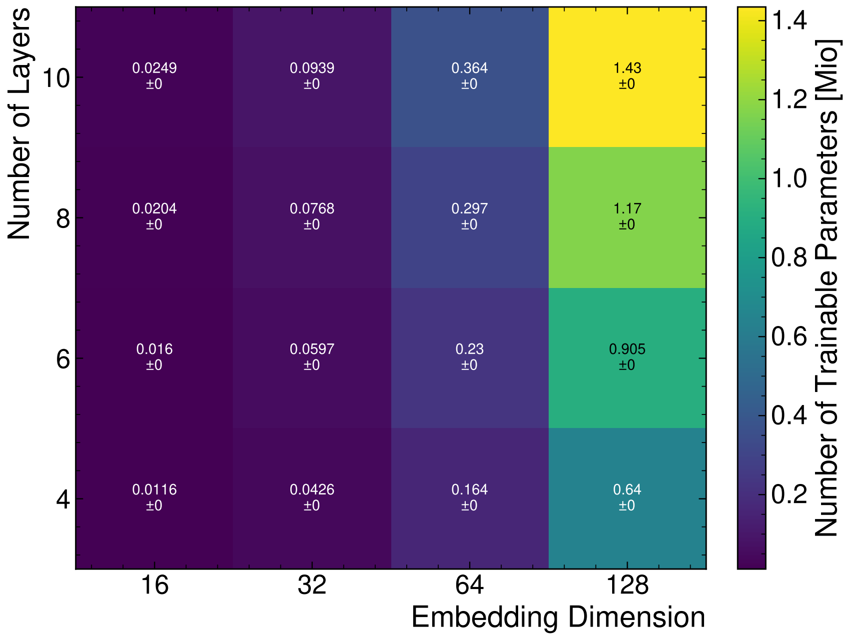
(b) Number of Trainable Parameters

Figure A.8.: The Figures (a) and (b) show the assignment accuracy and the number of trainable parameters of the FeatureConcatTransformer architecture as a function of the embedding dimension and the number of transformer layers.

A.2.2. CrossAttentionTransformer



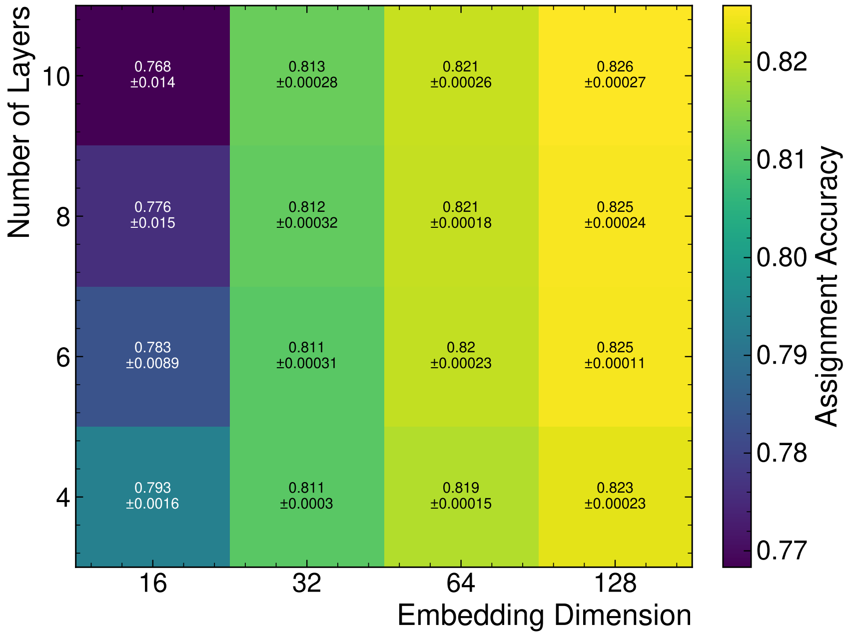
(a) Assignment Accuracy



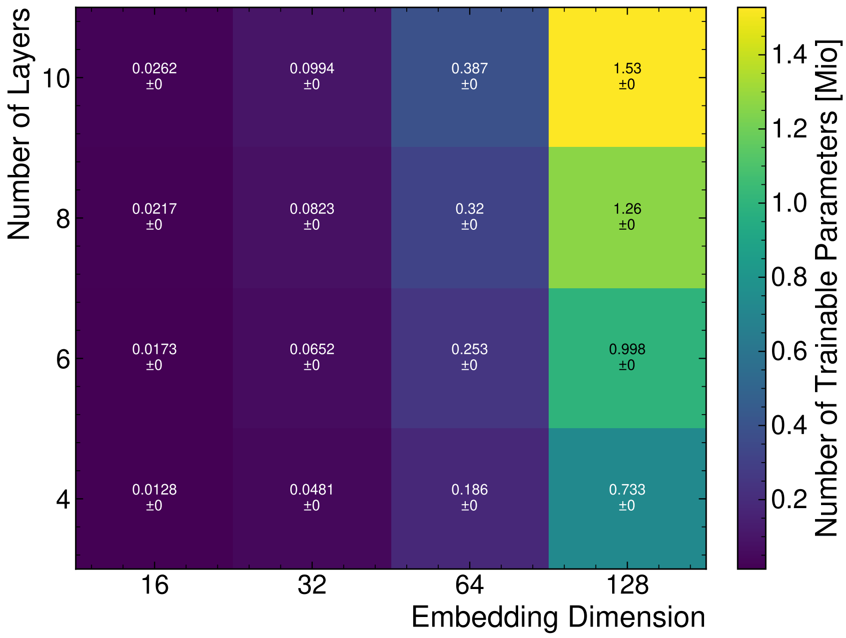
(b) Number of Trainable Parameters

Figure A.9.: The Figures (a) and (b) show the assignment accuracy and the number of trainable parameters of the CrossAttentionTransformer architecture as a function of the embedding dimension and the number of transformer layers.

A.2.3. CompactTransformer



(a) Assignment Accuracy



(b) Number of Trainable Parameters

Figure A.10.: The Figures (a) and (b) show the assignment accuracy and the number of trainable parameters of the CompactTransformer architecture as a function of the embedding dimension and the number of transformer layers.

A.3. Performance Plots

A.3.1. Assignment Performance

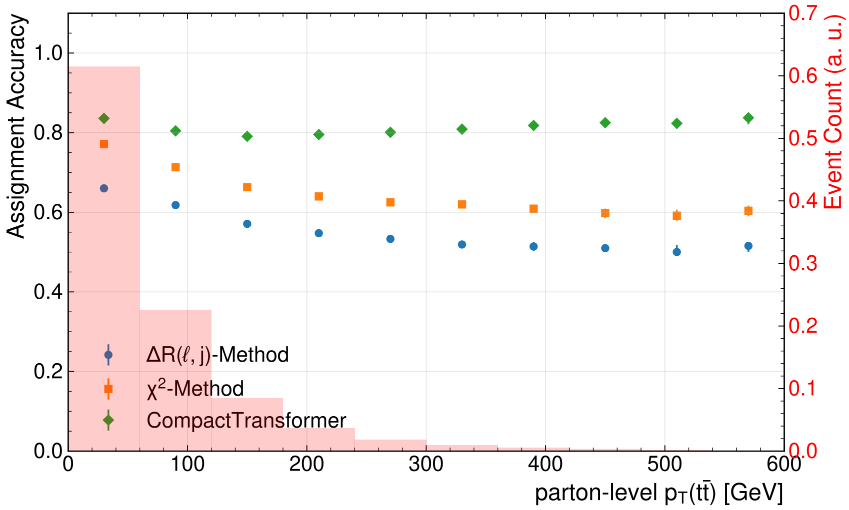


Figure A.11.: Assignment accuracy binned in the transverse momentum of top-quark pair system. The background histograms show of the distribution of events as a function of the number of jets. The uncertainties represent the statistical variations in the evaluation dataset.

A.3.2. Selection Performance

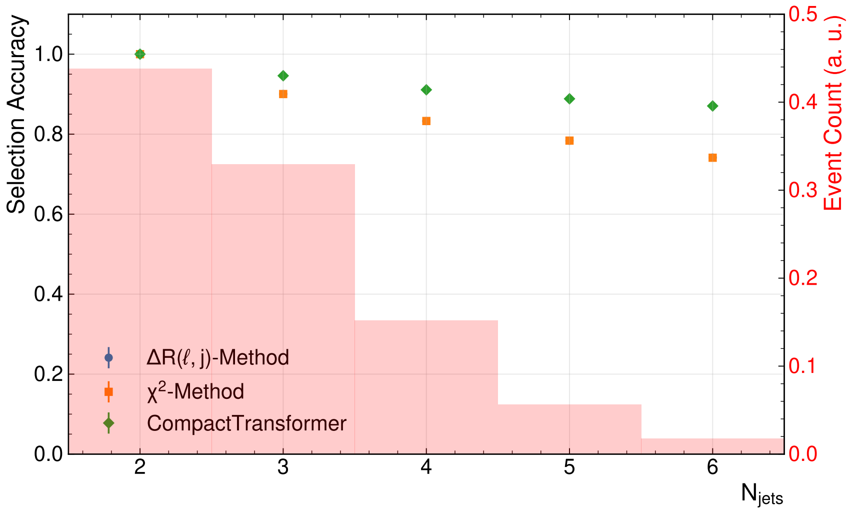


Figure A.14.: Selection accuracy as a function of the number of jets in the event for different reconstruction methods.

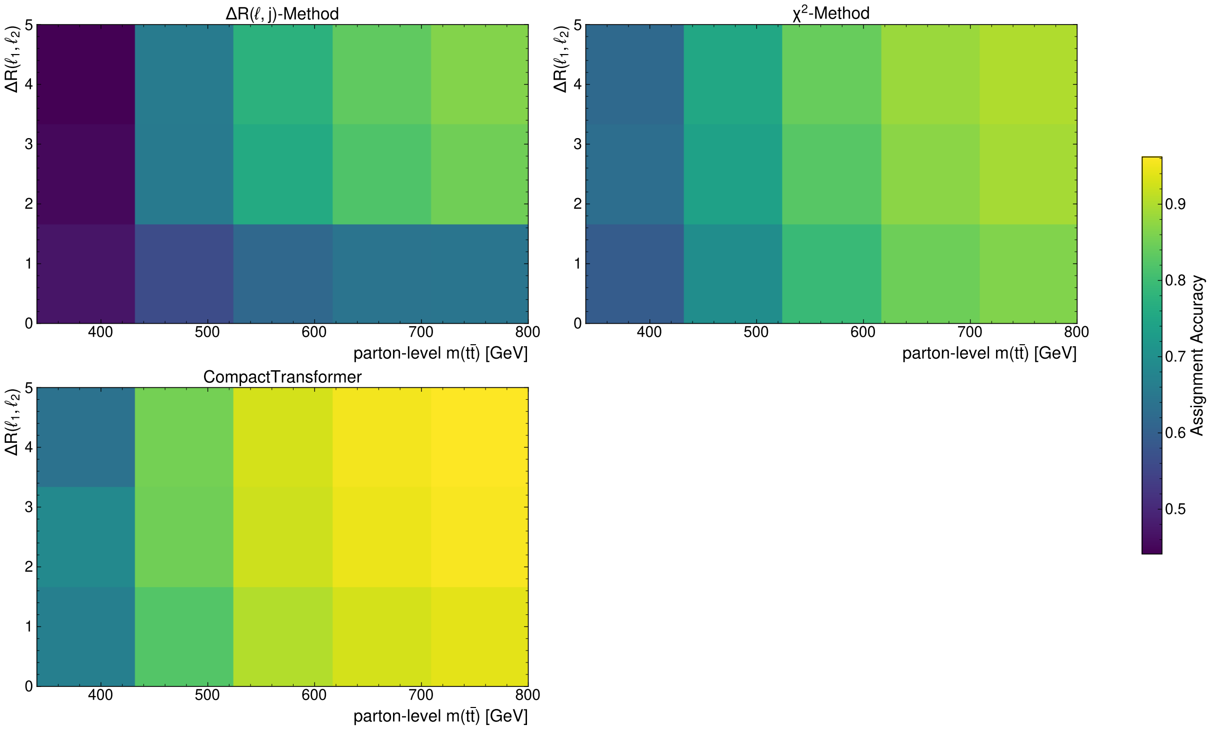


Figure A.12.: Assignment accuracy binned with respect to parton-level $m(t\bar{t})$ and $\Delta R(\ell_1, \ell_2)$.

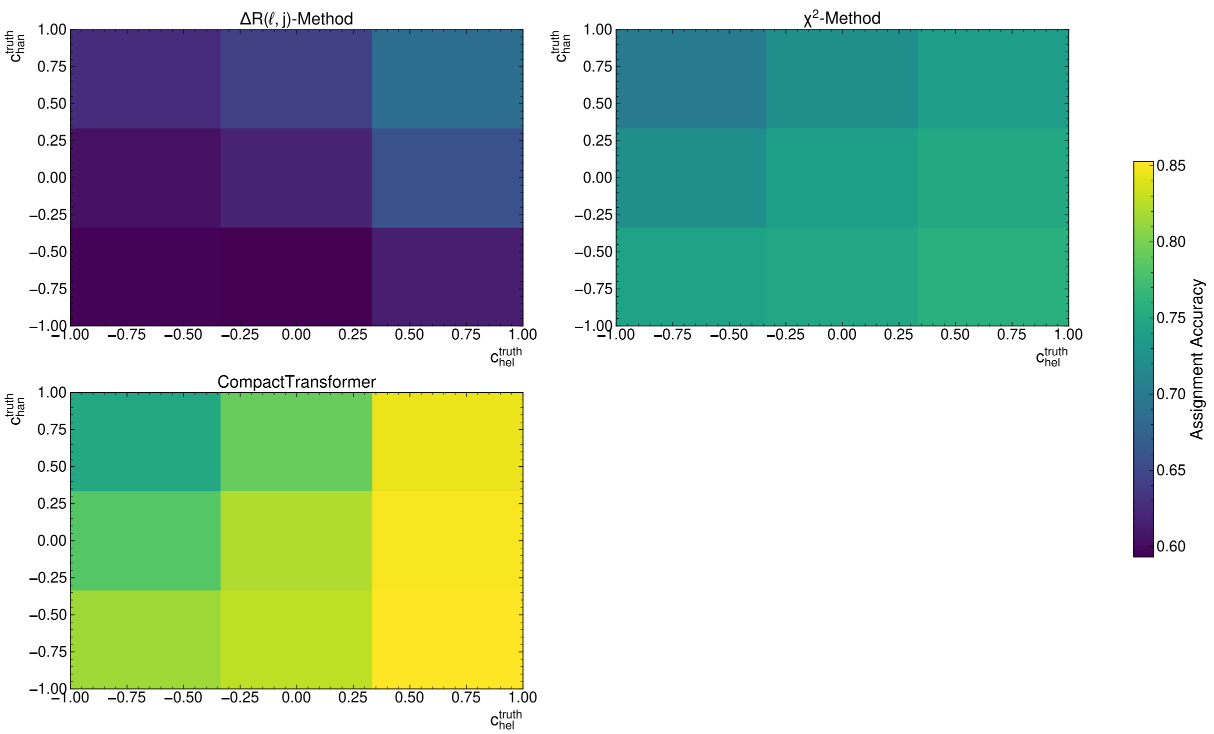


Figure A.13.: Assignment accuracy binned with respect to parton-level $c_{\mathrm{hel}}$and c_{han}.

B. Supplementary Computations and Proofs

B.1. Mathematical Foundation of Machine Learning

B.1.1. Mean Squared Error Loss and Conditional Expectation

When performing a minimisation of the mean squared error (MSE) loss function in a supervised learning setting, one can show that the optimal predictor corresponds to the conditional expectation of the target variable given the input features. The MSE loss function can be written as:

$$L_{\text{MSE}}(\mathbf{y}, f_\theta(\mathbf{x})) = \frac{1}{N}\sum_{i=1}^{N}(y_i - f_\theta(\mathbf{x}_i))^2 \tag{B.1}$$

Using a simple calculus argument, one can show that the function f_θ that minimises this loss function is given by:

$$\frac{d}{d\theta}L_{\text{MSE}}(\mathbf{y}, f_\theta(\mathbf{x})|\mathbf{x}) = \frac{d}{df}\int (y - f_\theta(x))^2 p(y|x)dy \cdot \frac{\partial f}{\partial \theta} \tag{B.2}$$

$$= \int 2(f_\theta(x) - y)p(y|x)dy \cdot \frac{\partial f}{\partial \theta} \stackrel{!}{=} 0 \tag{B.3}$$

$$\Rightarrow f_\theta(x) = \int yp(y|x)dy = \mathbb{E}[y|x] \tag{B.4}$$

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(Siemen Henning Aulich)